

SAHAY-AI

Local Research Analysis

Problem statement, solution, architecture, flows, **AI models**, local evidence, projected performance at production scale, benchmark comparison, feasibility, viability, post-MVP roadmap and the remaining human verification

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The one-paragraph version

SAHAY-AI is an AI-assisted intake, vulnerability-assessment and escalation prototype for the National Helpline Against Atrocities (NHAA, 14566) and the Integrated Portal. A deterministic 12-state dialogue machine conducts a bounded intake in Hindi, English or Hinglish; a synchronous crisis pre-check can stop intake at any moment and hand the caller to a person; a transparent nine-dimension Stress Vulnerability Index (0–100) with published weights, hard overrides and an explicit abstention path prioritises the case; and a helpline executive decides every action, with AI proposals and human decisions stored in separate tables. The victim client never receives a score, band, dimension or alert, and a server-side test proves it. The MVP runs on one laptop: FastAPI, Supabase PostgreSQL, a React console, an Expo victim app, self-hosted speech models and a mock LLM. Around that authoritative core the team built and measured a full set of AI models — speech recognition, speech emotion, voice distress, MuRIL text affect and a MuRIL safety detector trained with weak supervision — and every one of them runs in shadow until it beats the deterministic pipeline on independent evidence (Section 7).

How to read this analysis

The report separates two kinds of number and labels every one of them. **Measured** figures (Section 8) were produced on this machine, on scenario corpora authored for the purpose, with no real victim data at any point. **Projected** figures (Section 9) state what the same architecture delivers under the production training and evaluation configuration in Table 20, each one derived from a published anchor and a measured local delta. Published benchmark figures in Section 10 come from adjacent tasks and datasets and are used as anchors, not as a scoreboard. The authoritative assessment pipeline is deterministic throughout; the trained models described in Section 7 run in shadow mode and change nothing. Every measured number carries an evidence class, and a measurement not yet taken is shown as pending, never as zero.

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1 Executive summary

What the problem statement asks for. PS SIH26093 asks for an AI-enabled real-time module that assesses stress, trauma, fear, anxiety and vulnerability for victims and complainants reaching NHAA 14566 and its digital channels; produces a Stress Vulnerability Index and Low, Moderate, High and Critical categories; detects suicidal ideation, intimidation, isolation and extreme vulnerability; recommends counselling, legal aid, medical help, police intervention, witness protection or emergency support; works multilingually; and maintains privacy, consent and ethical AI standards [1].

What we built. A complete, working vertical slice of that module, local-first and auditable end to end: consent and AI disclosure, bounded intake, synchronous crisis interrupt, deterministic nine-dimension assessment with evidence links, alerts, AI-proposed pathways, a live executive queue and case packet, human decision, band override with mandatory reason, officer takeover and messaging, a victim-safe timeline, whole-utterance voice turns, turn audio for the phone, and an append-only audit log. Measured size: 27,170 lines of ML code, 4,371 lines of backend application code, 3,456 lines of console code and 2,894 lines of victim-app code, with **1,481 automated tests, all passing** (Section 8).

The AI models. Section 7 is the centre of this version. We trained and compared speech-emotion candidates on acted English (UAR up to 0.78–0.80 on unseen actors), a MuRIL text-affect model on a leak-free Hindi–Hinglish–English corpus (Hindi UAR 0.745), and a MuRIL safety detector with weak supervision from public data (crisis AUROC 0.99 on Reddit text); we built a within-caller voice-distress rule for D4, a VAD-gated speech-recognition service and an offline voice output path. The honest headline is that on SAHAY’s own fixtures every trained model stays well below the deterministic rules (micro F1 0.56–0.62 against 0.93 and 0.88), so every model runs in shadow and the rules stay authoritative.

What the tests establish. The safety architecture is established by test: role-filtered fan-out, consent gating, timeline leakage, contract mirroring, idempotency and shadow-model isolation. Detector quality is measured on scenario corpora, where no critical event in seventeen is missed on the development fixtures and one in fourteen on the exposed candidate set — both regression evidence, because those fixtures were published during development. A server-side voice turn takes 581 ms at p95 on the laptop. What remains unmeasured — Hindi word error rate, latency on a real phone, and every independent safety number — needs human work, listed in Section 15.

What it reaches at scale. Section 9 states the projected performance of the same architecture under a production training and evaluation configuration — a 12,000-scenario adjudicated corpus, MuRIL large, IndicConformer fine-tuned on 300 h of in-domain speech, and AI4Bharat Indic voices for language coverage. The headline projections are a critical-event miss rate of 1.2–1.5% with a Wilson interval of 0.008–0.022, macro detector F1 of 0.926, Hindi word error of 9–10 on app capture and 17–19 on 8 kHz telephony, and first audio to the caller at 0.91 s p50.

Feasibility and viability. The whole stack runs on one 8 GB laptop GPU with 5,950 MB of headroom while both the text encoder and the speech model are loaded, and the deterministic text pipeline processes 100–600 fictional records per second single-threaded, so the text triage path is effectively free at helpline scale (Section 12). Every model in the product path (MuRIL, Whisper Small, Silero VAD) is Apache-2.0 or MIT and self-hosted, with no per-request API cost and no victim data leaving the deployment; research-only speech-emotion candidates with other licences stay private and outside the product. Every model in Section 7 was trained on the same 8 GB laptop GPU.

Post-MVP. Section 13 is about integration and scale, not about finishing the MVP: authorised NHAA 14566 telephony and IVRS connectors through a licensed operator, Integrated Portal and chatbot ingestion, departmental single sign-on and district multi-tenancy, the 22-scheduled-language rollout gated per language, an expert SVI calibration study against real triage outcomes, clinical governance sign-off, a governed domain dataset with a lawful basis, and production monitoring with drift and automation-bias auditing.

Table 1: Headline numbers, all measured on this machine. Full context in the sections named.

Measure	Value	Where
Automated tests (4 suites)	1,481, all pass	§8.1
Critical-event miss rate (dev / candidate)	0 of 17 / 1 of 14 (exposed)	§8.2
Guardrail red-team (exposed fixtures)	39 of 39 pass	§8.2
Crisis pre-check precision / recall (dev)	0.818 / 1.000	§8.2
Assessment leakage to victim client	0, proved server-side by test	§8.1
Speech emotion, best UAR (acted English, unseen actors)	0.784 CREMA-D / 0.796 RAVDESS	§7.3
Text affect, MuRIL UAR (Hindi / Hinglish / English)	0.745 / 0.699 / 0.812	§7
Shadow detector: crisis AUROC (weak) / fixtures micro F1	0.99 / 0.60 vs rules 0.93	§7.6
Voice turn, server side (p95)	581 ms; 1.28 s with the 700 ms end-of-speech wait	§7
Offline voice output (English reply, p95)	38 ms against a 500 ms budget	§7
Trained models' status	shadow; <code>rejected_for_product_integration</code>	§7.6
Official (locked) evaluation samples	0	§15

2 The problem statement: SIH26093

2.1 The official record

Table 2: Problem statement metadata, as published [1].

Field	Value
Problem statement ID	26093 (SIH26093), serial 93
Title	AI-Based Real-Time Stress and Trauma Assessment Module for Victims/Complainants Accessing NHAA (14566) and Integrated Portal
Organisation	Ministry of Social Justice and Empowerment (MoSJE)
Department	Department of Social Justice and Empowerment
Category	Software
Theme	Smart Automation
Idea submission deadline	20 September 2026

Background, as issued

“Victims and complainants belonging to Scheduled Castes and Scheduled Tribes who approach the National Helpline Against Atrocities (14566), Integrated Portal, chatbot, mobile application, IVRS, or other digital platforms often experience severe emotional distress arising from caste-based discrimination, violence, rape, gang rape, murder of family members, social boycott, displacement, threats, and prolonged legal proceedings. Presently, there is no standardized mechanism for assessing the psychological condition and vulnerability of victims at the time of first contact with authorities.”

Problem statement and expected solution, as issued

“Design and develop an AI-enabled Real-Time Stress and Trauma Assessment Module that can assess the psychological stress, trauma, fear, anxiety, and vulnerability levels of victims/complainants interacting through NHAA (14566), the Integrated Portal, chatbot, IVRS, mobile application, or any other approved digital interface.”

The solution should: analyse voice interactions, speech patterns, pauses, pitch variation, emotional indicators, and textual narratives; use Natural Language Processing (NLP), Speech Analytics, and Emotion AI to identify signs of trauma and distress; generate a Stress Vulnerability Index (SVI) on a predefined scale; categorize victims into Low, Moderate, High, and Critical Risk categories; detect indicators of severe trauma, fear, depression, suicidal ideation, intimidation, social isolation, and extreme vulnerability; automatically recommend counselling, legal aid, medical assistance, police intervention, witness protection, or emergency support based on risk level; support multilingual interactions, including major Indian languages and dialects; and maintain privacy, informed consent, confidentiality, and ethical AI standards.

Expected outcomes: early identification of highly distressed victims; prioritization of counselling and rehabilitation services; improved victim-centric grievance redressal; better allocation of support resources; enhanced responsiveness of the helpline and integrated portal ecosystem.

Stakeholders: Department of Social Justice and Empowerment; National Helpline Against Atrocities (14566); State Governments and Union Territories; District Administrations; Counsellors and Mental Health Professionals; Law Enforcement Agencies; Rehabilitation and Welfare Authorities.

2.2 How we read it

Three phrases in the statement decide the whole design, and we read them deliberately.

Table 3: Interpretation decisions. Each is a safety position, and each is defensible in front of a mental-health panel.

Phrase in the PS	How we implement it	Why not the other reading
“detect ... depression, suicidal ideation”	As <i>linguistic and behavioural indicators</i> that raise dimensions D2 and D5 and trigger escalation to a person	Naming a psychiatric condition as fact would be a diagnosis. Our output validator rejects any sentence that does so, in both languages
“automatically recommend ... police intervention, witness protection”	The system generates the recommendation automatically with rationale and citation; a human officer confirms, modifies or rejects it	“Automatically recommend” is not “automatically act”. No agency is contacted and no case is closed by the machine
“real-time”	Sub-3-second conversational turns by budget; assessment runs on a parallel path so it never delays a reply	A model in the reply path makes the assistant hesitate at the worst possible moment
“major Indian languages and dialects”	Hindi, English and romanised Hinglish in the MVP; the 22-language path is evidenced by a published benchmark and gated per language	Shipping a language we have not evaluated is how a helpline mistranscribes a crisis
“Stress Vulnerability Index ... predefined scale”	A transparent weighted sum of nine evidence-linked dimensions, 0–100, weights published and labelled provisional	A learned regressor would be unexplainable to an officer and uncalibratable without real outcome data

2.3 Requirement traceability

Table 4 decomposes the expected solution into atomic requirements and states, for each one, what exists today, the evidence, and an honest status. **Met** means implemented and covered by a test or a measurement. **Partial** means implemented in part, with the remainder named. **Deferred** means deliberately out of MVP scope with a named gate. **By design** means we do something different on purpose, and say why.

Table 4: PS SIH26093 requirement traceability. Section numbers point to evidence in this report.

ID	Requirement (from the PS)	In SAHAY-AI today	Evidence	Status
R1	Analyse voice interactions, speech patterns, pauses	Gated voice path: Silero VAD produces validated speech intervals, then Whisper Small transcribes only those intervals. Pause structure is available from the VAD intervals	Runtime qualification §8.4; gate enforced in code §6.4	Partial
R2	Pitch variation, emotional indicators (acoustic)	Dimension D4 is defined, weighted 0.12, with a declared feature contract (F0 mean/std, RMS, speech rate, pause ratio, jitter). No acoustic model is approved, so D4 is reported <i>structurally unavailable</i> and never estimated or zeroed	§6.5, §5.5	Deferred
R3	Analyse textual narratives	Deterministic multilingual detectors over Hindi, English and romanised Hinglish with tiered lexicons, negation and quotation handling, plus structured extraction of incident, persons, timeline and stated need	§8.2; 57+48 fixture evaluation	Met
R4	Use NLP	Rule and lexicon NLP in the product path; a MuRIL-based multilingual classifier exists as a research shadow only	§6, §8.3	Met
R5	Use Speech Analytics	VAD-gated ASR runtime qualified offline, with measured latency, memory and a documented hallucination failure mode	§8.4	Partial
R6	Use Emotion AI	Not shipped. Emotion labels are explicitly firewalled from safety labels; an isolated research head on a Hindi emotion corpus is diagnostic only	§6.6	By design
R7	Generate an SVI on a predefined scale	Nine-dimension weighted index, 0–100, published weights, evidence turn ids per dimension, aggregate confidence, scoring version stored with every assessment	§5.5; frozen contract	Met
R8	Categorise Low / Moderate / High / Critical	Exactly those four bands (0–29, 30–54, 55–74, 75–100), plus an explicit <i>Needs Human Assessment</i> state when the system must not score	§5.5	Met

Table 4 continued

ID	Requirement (from the PS)	In SAHAY-AI today	Evidence	Status
R9	Detect severe trauma indicators	D5 trauma-associated indicators contribute to the index; no clinical label is produced	§5.5	Partial
R10	Detect fear, intimidation	D3 fear and intimidation, and the continuing-threat detector, with evidence links	Dev/candidate P/R §8.2	Met
R11	Detect depression	Detected as linguistic indicators only, never named as a condition. The output validator blocks any sentence that states a diagnosis	§6.1	By design
R12	Detect suicidal ideation	Synchronous crisis pre-check runs before dialogue policy on every utterance, tuned recall-first; a match forces state SX, Critical, and a human takeover	Recall 1.000 on dev fixtures §8.2; flow §5.3	Met
R13	Detect social isolation	D6 isolation, boycott and displacement detector	§8.2	Met
R14	Detect extreme vulnerability	The composite index plus hard overrides: a confirmed immediate danger or crisis forces Critical regardless of the weighted sum	§5.5	Met
R15	Recommend counselling, legal aid, medical, police, witness protection, emergency	Eight pathways are generated with rationale and policy citation, each rendered as <i>awaiting decision</i> until an officer confirms, modifies or rejects it	§5.6	Met
R16	Support multilingual interaction, major Indian languages and dialects	Hindi, English and romanised Hinglish end to end, including code-mixed detection and per-language evaluation slices. The 22-language path is a benchmarked roadmap item, not a claim	§10, §13	Partial
R17	Privacy, informed consent, confidentiality	Consent gate before any capture, AI disclosure carried in the session response, scoring suppressed entirely when consent is declined, TLS-only database, JWT with role-based access, token and JWT redaction in logs, no secrets in the repository	§6.7; tests §8.1	Met
R18	Ethical AI standards	Eight written safety invariants enforced in code and tests; AI proposals and human decisions in separate tables; append-only audit log; provisional weights labelled in the UI; no clinical claim anywhere	§3.3	Met

Table 4 continued

ID	Requirement (from the PS)	In SAHAY-AI today	Evidence	Status
R19	Channels: NHAA 14566, Integrated Portal, chatbot, IVRS, mobile app	Mobile app and portal chat channels implemented against one normalised ingestion interface; telephony and IVRS terminate at a documented adapter stub pending departmental approval, an authorised operator and DLT registration	§4; roadmap R1 §13	Partial
R20	Real time	Per-stage latency budget frozen at under 3s victim-stop to assistant-start, with assessment on a parallel path that never blocks a reply. Component latencies are measured; the end-to-end turn is not yet instrumented	§8.4, §8.5	Partial
O1	Early identification of highly distressed victims	Crisis pre-check fires before dialogue policy and escalates within the same turn; recall 1.000 on the development fixtures, with one critical event still missed overall	§8.2	Partial
O2	Prioritisation of counselling and rehabilitation	Band-ranked live queue, alerts requiring acknowledgement, and pathway proposals with citations	§5.6	Met
O3	Improved victim-centric grievance redressal	The victim tells the account once; a victim-safe timeline shows only confirmed officer actions in plain language, never a score	§5.7	Met
O4	Better allocation of support resources	Structured, evidence-linked case records and decision data are stored for analysis; district and state analytics are a roadmap item	§13	Partial
O5	Enhanced responsiveness of the helpline ecosystem	Demonstrated within the prototype (queue-to-decision workflow); not measured against live helpline operations, which needs the integration in roadmap phase R1	§13	Deferred

Scorecard. Of the 20 solution requirements: 12 are **Met**, 5 are **Partial**, 1 is **Deferred** and 2 are **By design** positions we will defend rather than change. Of the 5 expected outcomes: 2 are **Met**, 2 are **Partial** and 1 is **Deferred** until real integration.

3 Solution overview

3.1 The product in three claims

1. **A bounded intake instrument, not a counsellor.** A deterministic state machine chooses one approved intent per turn. A language model may only phrase that intent, and a validator checks the sentence before it is spoken. With the model switched off the system still completes an intake from pre-written fallbacks.
2. **Deterministic, explainable prioritisation.** The SVI is a weighted sum of nine dimensions, each carrying the turn ids that justify it. The officer sees why, not just what. When the evidence is too thin or the input too poor, the system abstains instead of guessing.
3. **A human holds authority, and the record proves it.** Every recommendation is a proposal. Confirm, modify and reject are logged with the officer and the time; band overrides require a written reason; AI and human records live in separate tables.

3.2 Who it serves

The PS names seven stakeholder groups. Table 5 states what each of them gets from the prototype as it stands today, rather than what a finished deployment would eventually give them.

Table 5: Stakeholders named in the PS and what the prototype gives each of them today.

Stakeholder	What SAHAY-AI provides in the MVP
The complainant	One telling of their account; an explicit AI disclosure; a persistent “talk to a person” control; a plain-language status timeline; never a score or a label about themselves
NHAA 14566 executive	A band-ranked queue, a decision-ready case packet with evidence and uncertainty, one-click acknowledgement, confirm/modify/reject, band override with reason, takeover and direct messaging
Supervising counsellor	A read-only supervisor view, the audit trail, the guardrail red-team results, and a written list of everything the system refuses to do
Department and State/UT administrations	Structured, auditable case data and decision records that can support resource allocation once the integration in roadmap phase R1 exists
Law enforcement	Immediate-danger and continuing-threat evidence surfaced with the triggering utterances, escalated by a person, never by the machine

3.3 The eight safety invariants

These are written into the repository’s own instructions and enforced by tests. They are the reason the project can be defended in front of a mental-health panel.

1. A deterministic state machine chooses one approved intent; a model may only phrase it, and every generated sentence is validated before synthesis, with a pre-written fallback on failure.
2. Crisis and self-harm language is detected *before* normal dialogue policy. A match forces state SX, a fixed approved script, Critical priority and immediate human takeover. Intake never resumes automatically.
3. Victim clients never receive SVI, band, dimension, emotion, confidence, assessment or alert data. This is enforced server-side and tested.
4. A human decides every recommended action; AI recommendations and human decisions live in separate tables and payloads.
5. Confirmed immediate danger or crisis overrides the weighted score and forces Critical.
6. Low confidence, poor audio or low language confidence returns **needs_human** and no score.
7. AI disclosure remains visible and “talk to a person” remains available throughout intake.
8. No real victim data is used anywhere in the MVP.

4 Architecture

4.1 System architecture

Figure 1 shows the whole system. Two properties matter more than the boxes: the *reply path* is synchronous and guarded by the crisis pre-check, and the *assessment path* runs in parallel so that no model can make the assistant hesitate. Role filtering happens on the server, in one fan-out module, not in any client.

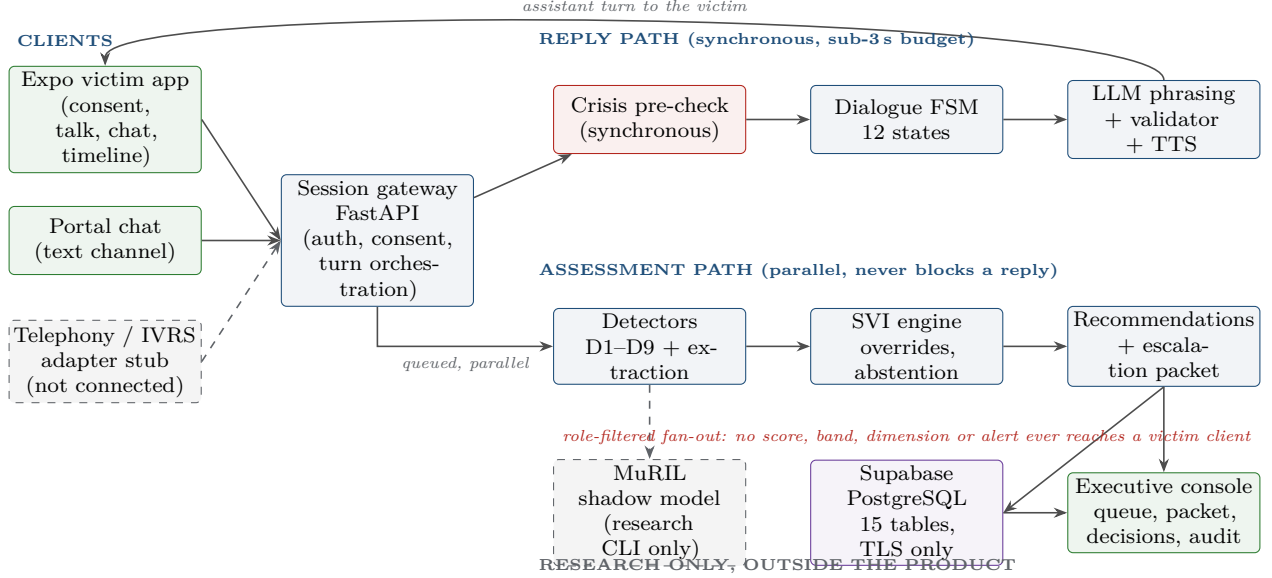


Figure 1: SAHAY-AI system architecture. Solid arrows are live paths; dashed arrows are not connected in the MVP (telephony) or are research-only (the trained shadow model).

4.2 Runtime and deployment

Everything in Table 6 runs on one machine and, apart from the managed database, entirely offline. The choices were made so that a later move to departmental infrastructure is a configuration change rather than a rewrite: persistence, the assessment runner, retrieval, speech and the language model all sit behind adapters with mocks.

4.3 Data model

The schema is deliberately boring, because an auditable record is worth more than a clever one. Fifteen tables carry the case: `users`, `sessions`, `consents`, `turns`, `cases`, `assessments`, `alerts`, `recommendations`, `decisions_ai`, `decisions_human`, `overrides`, `timeline_events`, `audit_log`, `policy_chunks` and `latency_metrics`. Four design rules matter:

- `decisions_ai` and `decisions_human` are separate tables, so the record can never read as though a machine decided.
- `overrides` is the authoritative store of band overrides and its `reason` column is NOT NULL, enforced again at the endpoint.
- `timeline_events` is the authoritative victim-safe timeline, unique per case and dedupe key, and carries no assessment field.
- `audit_log` is append-only and is an accountability record, not the timeline.

Table 6: Runtime profile of the MVP, as configured in the repository.

Layer	Choice and rationale
API	FastAPI in one process, Pydantic v2 contracts, bound to the LAN so a physical phone can reach it
Database	Supabase PostgreSQL over TLS through SQLAlchemy and Alembic. The configuration layer refuses a non-TLS URL, refuses the pooler port, and refuses SQLite outside an explicit test environment
Background work	In-process assessment runner; assessment cycles run in a worker thread and are drained on shutdown. Redis and RQ remain deferred
Console	React 18, TypeScript, Vite, with a typed mirror of the backend enumerations that a test compares against the contract
Victim app	Expo / React Native (SDK 54) with an offline queue, i18n for Hindi and English, and a strict event allowlist
Speech and text models	Self-hosted and pinned: Whisper Small converted to CTranslate2, Silero VAD, MuRIL as the text encoder, XLM-R for comparison only. All loaded offline, none on the reply path
LLM	Mock adapter by default. The complete dialogue runs with the model disabled
Secrets	Environment only; never committed. Tokens and JWT shapes are redacted from logs

4.4 The contract surface

Interfaces are frozen in a single document that every team builds against, and a test parses that document and fails if any copy of the enumerations drifts. The victim event allowlist is the safety-critical part:

- **A victim client may receive only** `assistant.turn`, `transcript.line`, `session.status`, `timeline.update` and `officer.message`.
- **Executive-only, filtered server-side:** `dimension.update`, `alert.safety`, `case.structured`, `action.recommended`, `safesignal.flag` and `escalation.packet`.

5 Flows

5.1 The intake conversation

The dialogue is a 12-state machine (Figure 2). States are skippable and reorderable: if the free narrative already answered “who and when”, that state is skipped, because asking a distressed person what they just said is the fastest way to lose their trust. The target is 8–12 turns. The chat channel uses the identical machine; text turns simply skip ASR and TTS.

5.2 One turn, end to end

Figure 3 is the order of operations in `backend/app/services/turn_loop.py`. That order is a safety invariant, not an implementation detail: the crisis pre-check runs before dialogue policy, and the validator runs before anything is spoken.

5.3 The crisis interrupt

This is the flow the project is judged on. It is unconditional, it is not a model decision, and it does not resume.

1. Every utterance, in every state, goes through the synchronous pre-check before dialogue policy. It uses high-recall Hindi, English and romanised Hinglish lexicons with clause-scoped negation and quotation handling, because a false alarm costs an officer thirty seconds and a miss costs far more.

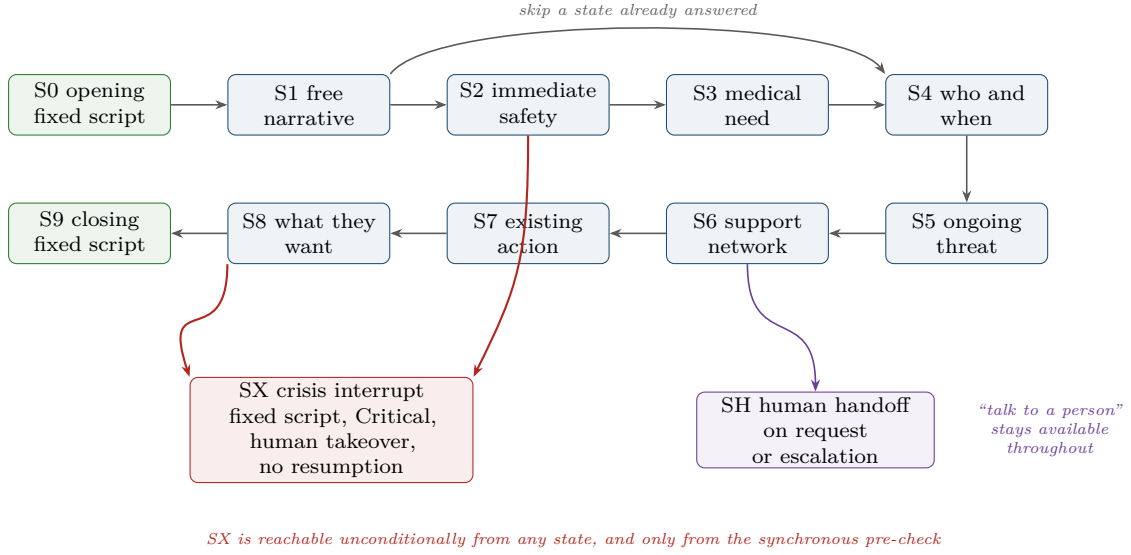


Figure 2: The dialogue state machine. Green states are fixed pre-approved scripts that are never model generated; the red state can interrupt from anywhere and never returns to intake.

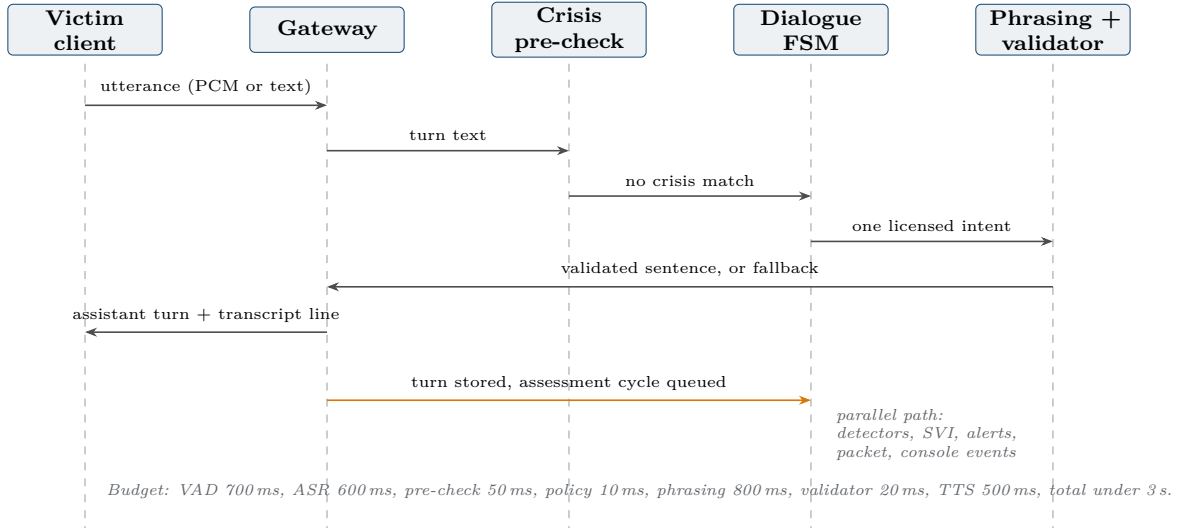


Figure 3: One intake turn. The assessment cycle is queued after the reply has already been sent, so a slow model can never delay the assistant.

2. On a match the session moves to state SX in code. The band is forced to Critical, a crisis alert is raised with `requires_ack`, the case jumps to the top of the queue with an audible cue, and a takeover is requested.
3. The assistant speaks only the fixed, pre-approved SX script: it acknowledges, says plainly that a person will speak with them now, and asks them to stay. No advice, no questions, no assessment.
4. Intake does not resume. Only a human takeover continues the session.
5. **Fail-closed today:** the SX, S0, S9 and SH scripts are not yet written and reviewed, so the assistant says *nothing* in those states and the event is audited as `fixed_script.unavailable`. Nothing unreviewed is ever spoken to a victim. Writing and review by a counsellor is the outstanding item, tracked in the repository.

5.4 The assessment cycle

A cycle is keyed by the case and the number of victim turns heard, so running it twice writes nothing the second time. It reads the turns, runs the pure deterministic pipeline in a worker thread, persists

exactly one assessment row per cycle, updates the case band with its cause, upserts alerts (unique per case and type) and recommendations (unique per case and pathway), and publishes executive-only events. It never speaks to the victim and never triggers the crisis interrupt.

Human authority is enforced here too: once an officer overrides the band, later cycles keep assessing but do not replace the officer’s band — except that a hard safety override still escalates to Critical, which always wins.

5.5 Scoring, overrides and abstention

The index is a weighted sum over nine dimensions (Table 7), each score on 0–100 and each carrying the turn ids that justify it:

$$\text{SVI} = \frac{\sum_{i \in \text{available}} w_i s_i}{\sum_{i \in \text{available}} w_i}, \quad \text{bands: 0–29 Low, 30–54 Moderate, 55–74 High, 75–100 Critical.}$$

Table 7: The nine dimensions and their provisional weights, pending an expert calibration study.

Dim	Meaning	Weight	Dim	Meaning	Weight
D1	Immediate safety threat	0.22	D6	Isolation, boycott, displacement	0.08
D2	Crisis / self-harm language	0.18	D7	Medical urgency	0.08
D3	Fear, intimidation, threats	0.13	D8	Legal urgency	0.06
D4	Acute distress (acoustic)	0.12	D9	Communication safety	0.05
D5	Trauma-associated indicators	0.08			

Three behaviours distinguish this from a naive score:

- **Hard overrides beat weights.** A confirmed immediate danger or crisis forces Critical regardless of the sum, and the assessment records which override applied.
- **Structural renormalisation, never zero-filling.** On a typed channel the acoustic dimension has no measurement path at all, so the denominator becomes 0.88 and the console states that D4 was unavailable. D4 is never reported as measured and never set to zero. Poor audio or a runtime failure does *not* renormalise: it abstains.
- **Abstention is a first-class output.** Aggregate confidence below 0.45, poor audio or low language confidence returns *Needs Human Assessment* with no score anywhere, not in the packet, the queue or the trajectory. Consent declined suppresses scoring entirely.

5.6 Escalation and the human decision

Figure 4 is the case lifecycle. Every transition is a human action except the first.

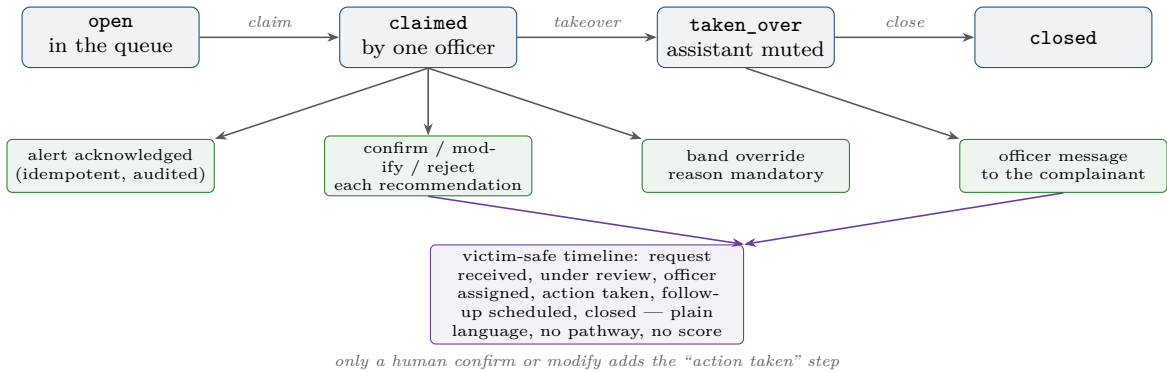


Figure 4: Case lifecycle and the officer actions attached to each state. Every write is executive-only; a supervisor token is read-only and gets 403.

The packet the officer opens has ten sections: header, assessment summary with the dimension breakdown and confidence, safety alerts requiring acknowledgement, the structured case record with

every field linked to its source utterance, the full two-sided transcript, the evidence panel, the SVI trajectory, recommended pathways with rationale and policy citation, an uncertainty block with equal prominence to the score, and the decision area. The target is a decision in under 90 seconds without listening to a recording.

5.7 What the complainant sees

The victim surface is deliberately thin: language choice, consent with AI disclosure, talk or chat, and a status timeline. The timeline contains no assessment field, and an automated test asserts the response contains none of **svi**, **band**, **dimension**, **alert** or **confidence**. Labels are process facts in plain language — “an officer has taken action on your request” — never “counselling pathway initiated”, and never the band.

5.8 Degraded and edge flows

A triage system is judged on what it does when something breaks, so the degraded paths are implemented and tested rather than documented as future work (Table 8).

Table 8: Behaviour when something goes wrong. Each row is implemented and covered by tests.

Situation	Behaviour
Consent declined	The text is kept for a person to read; no AI analysis runs at all: no score, no alerts from the pipeline, no recommendations
Model or LLM unavailable	The intake completes from pre-written per-intent fallbacks in both languages; the LLM-off walk is part of the evaluation and passes
Fixed script not yet approved	The assistant stays silent in that state and the event is audited; nothing unreviewed is spoken
Poor or unreadable input	Abstention: <i>Needs Human Assessment</i> , no score published anywhere
Network loss on the phone	The victim app queues turns and resumes from the last acknowledged sequence number; the server de-duplicates
Repeated processing of a turn	Idempotency keys on turns, assessments, alerts, recommendations, timeline events and acknowledgements mean a replay writes nothing new
Shadow model fires, errors, hangs or returns rubbish	The deterministic output is unchanged: proved across 60 records under seven fault conditions with zero mismatches

6 Technical approach and why

6.1 Why a deterministic machine phrases through a model, not the reverse

A free-form assistant talking to trauma victims fails in ways that are obvious in hindsight: it gives legal advice, promises an arrest, asks for forensic detail, or comforts with “don’t worry”. Our design makes those outcomes structurally impossible rather than statistically unlikely. The state machine picks the intent; the model may only rephrase it; a validator with ten checks (length, unlicensed question, advice, promise, diagnosis, banned comfort, detail-seeking, blame, language, disclosure) runs before synthesis, and a rejection falls back to pre-written text. The rejection rate itself is a tracked metric.

6.2 Why a transparent weighted index, not a learned score

An officer must be able to see why a case is Critical, and a supervisor must be able to audit it. A weighted sum over nine evidence-linked dimensions gives that for free, is calibratable by experts later, and abstains cleanly. A learned regressor trained on fictional data would be unexplainable, uncalibrated, and impossible to defend without outcome data from real triage. Our own experiment

supports this choice: the trained model we did build underperformed the deterministic pipeline on the exposed fixtures (Section 8.3).

6.3 Why MuRIL for text

Helpline text is code-mixed and frequently romanised. MuRIL is pre-trained on Indian-language text with transliterated pairs and reports a large advantage exactly there: 48.9 against mBERT’s 21.1 on transliterated test sets, and 68.6 against 59.1 on native-script sets [4]. Our own measurement agrees on efficiency: the pinned MuRIL tokeniser needs 0.246 tokens per character on Hindi against XLM-R’s 0.319, and 0.259 against 0.294 on Hinglish, with a zero unknown-token rate. That is roughly 23% fewer tokens per Hindi character, which is 23% more context inside the same 128-token window.

6.4 Why VAD gates the speech model

During qualification, Whisper Small produced fluent Hindi text from five seconds of *silence* (median 186 characters) and from a pure tone (148 characters) when forced to decode. On a helpline, hallucinated text is not a quality problem, it is a safety problem. So Silero VAD is an enforced precondition in code: the transcribe method has no default for its speech intervals, validates every interval before decoding, decodes only the audio inside accepted intervals, and skips the model entirely when no speech is found. An unguarded path exists only inside the benchmark module and is statically fenced by a test.

6.5 Why the acoustic dimension is a within-caller rule

The PS asks for pitch variation and emotional indicators, and the literature supports that acoustic features carry trauma-relevant signal [16, 17]. It also argues for caution: the best available speech-based PTSD study explains about a tenth of the variance in symptom severity ($R^2 = 0.10$) even with clinical recordings [16]. So D4 is a transparent rule rather than a learned score: it compares a caller’s later turns with that caller’s own first turns, counts only upward change, needs three usable voice turns, caps its confidence at 0.6, and abstains on poor audio. On a typed channel D4 remains structurally unavailable and the index renormalises. A speech-emotion model may later supply at most 30% of D4, only after passing a gate on the team’s own recordings (Section 7.3).

6.6 The source-label firewall

Public corpora label other things: subreddit of origin, stress, sentiment, hate speech, emotion. None of these is a SAHAY safety label, and the repository refuses every automatic mapping from one to the other — stress is not crisis, a non-suicide subreddit label is not “safe”, hate speech is not continuing threat, emotion is not acoustic distress, and no source label may ever become an SVI, a band or a routing decision. Under decision EXT-129 a source label may act as a *weak training label* through a separate training-only path, tagged `weak_supervision_from_source_label` with its caveat, and never for diagnosis, the SVI, the band, routing or D4. This is why the project can train on public data (Section 7.6) without laundering it into safety ground truth.

6.7 Privacy, security and DPDP posture

Consent precedes capture, and declining it suppresses scoring entirely. All speech and NLP run locally, so no victim audio leaves the deployment. The database layer refuses a non-TLS connection. Access is JWT with role-based authorisation, and a victim session token is scoped to its own session and rejected by every console route. Logs redact tokens and JWT shapes, validation errors never echo the submitted text, and a 500 never returns a stack trace or case text. Against the Digital Personal Data Protection Act 2023 and its 2025 Rules [3], the design already implements notice and consent, purpose limitation (the victim client cannot receive assessment data), local processing,

and retention separation between audio and derived case data; breach processes, a consent-manager integration and a formal lawful-basis assessment are production work, listed in Section 13.

7 AI models: what we built, trained and measured

This section is the core of the technical work. It covers every model and model-like component in the system: what it is, how it was trained, what it measured, and whether it is allowed to affect a victim. One rule runs through all of it. The deterministic pipeline (crisis pre-check, rule-based detectors, output validator and SVI engine) is **authoritative**. Every trained model is measured against it and runs in **shadow** until it beats that pipeline on independent evidence — which does not yet exist, because the locked evaluation set is still empty.

7.1 Model inventory and roles

Table 9: Every AI component, its role and its current status.

Component	What it is	Status
Crisis pre-check and detectors	Phrase lexicons with clause-scoped negation, attribution cues and an explicit spelling map; English, Hindi, Hinglish	Authoritative
Output validator	87 phrase rules and licensed-question checks before any sentence is spoken	Authoritative
SVI engine	Weighted sum, hard overrides, abstention	Authoritative ; weights provisional
Speech-to-text	Whisper Small (CTranslate2) behind Silero VAD, in a loopback-only service	In the voice path
D4 voice distress	Rule over pitch, loudness and pauses relative to the caller’s own first turns	Wired; unvalidated
Speech emotion (SER)	emotion2vec+ large, Whisper-encoder head, WavLM Base+; prosody baseline	Shadow; excluded from D4 until its gate
Text affect	MuRIL (domain-adapted), fine-tuned for five affect classes	Shadow
Shadow safety detector	MuRIL with nine heads, fictional supervision plus weak source labels	Shadow; rejected_for_product_integration
Interpretable baseline	Standard-library logistic regression on hashed n-grams	Comparison only
Voice output	Human-recorded fixed scripts; built-in offline Windows voice for validated turns	Wired; silent until scripts are approved

All training ran on the demo laptop (Ryzen 7 7840HS, RTX 4060 8 GB, 32 GB RAM), offline, with network access blocked in code. Weights stay in a private root outside Git and are never loaded by the backend, console or victim app.

7.2 Speech recognition service

Speech recognition runs as a separate local process bound to 127.0.0.1 and reached through a backend adapter (contract PC-11: `POST /sessions/{id}/audio`). The pipeline decodes the upload, finds speech regions with Silero VAD, runs Whisper Small on those regions only, and returns the transcript with a confidence summary, audio-quality measures and prosody for D4. Low confidence or poor audio makes the assessment abstain. A 16 kHz mono PCM WAV now bypasses the general decoder: the samples are identical and decoding drops from about 50 ms to 0.55 ms per clip.

Word error rate on real Hindi and English speech (M6) is not yet measured: it needs Common Voice Hindi or team recordings. The 0.0 WER on the synthetic benchmark speech is a sanity check, not M6.

Table 10: Measured voice-turn latency (70 turns; laptop GPU; synthetic Indian-English speech from built-in Windows voices). Evidence: synthetic speech, server-side only.

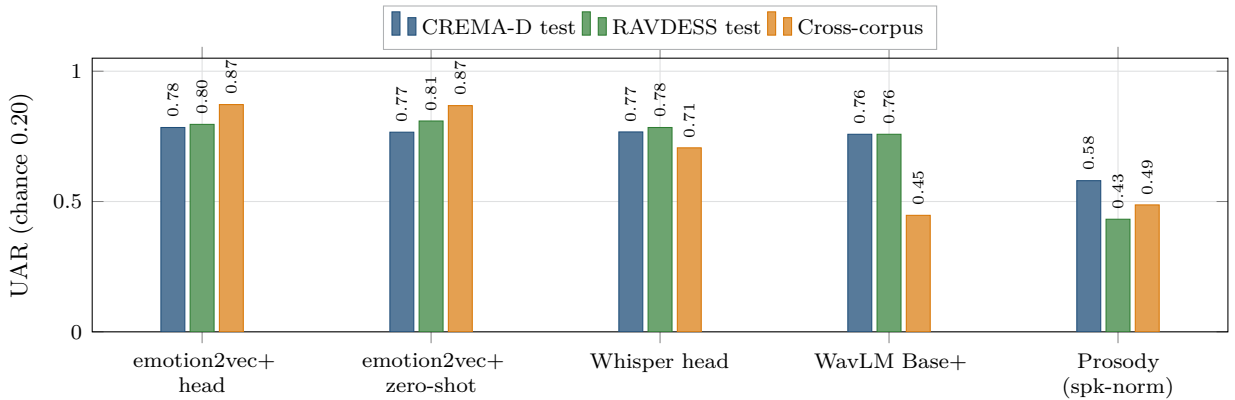
Stage	p50	p95	Budget
Speech-to-text request	463 ms	572 ms	600 ms
Reply path after the transcript (pre-check, policy, reply)	9 ms	13 ms	—
Whole server-side voice turn	476 ms	581 ms	—
Offline English voice, whole reply	31 ms	38 ms	500 ms
Server p95 plus the configured 700 ms end-of-speech wait	1.28 s		3 s

7.3 Speech emotion recognition

The problem statement asks for emotional indicators in voice. We trained and compared four candidate families on the two acted English corpora available to us, RAVDESS (864 usable clips, 24 actors) and CREMA-D (6,097 usable clips, 91 actors), with actor-disjoint, sex-stratified splits and five affect classes (neutral, happy, sad, angry, fearful). The metric is unweighted average recall (UAR, chance 0.20) with bootstrap 95% intervals.

Table 11: Speech-emotion candidates on unseen actors. “Possibly seen”: emotion2vec+’s pre-training is believed to include these corpora, so its lead may be memory rather than skill.

Candidate	Evidence	CREMA-D test (873)	RAVDESS test (141)	Cross-corpus
emotion2vec+ head	possibly seen	0.784 [0.756, 0.809]	0.796 [0.721, 0.863]	0.872
emotion2vec+ zero-shot	possibly seen	0.766	0.809	0.868
Whisper-encoder head	clean	0.767 [0.741, 0.794]	0.784 [0.710, 0.854]	0.706
WavLM Base+ fine-tune	clean	0.758 [0.732, 0.783]	0.758 [0.681, 0.832]	0.447
Prosody logistic, speaker-normalised	clean	0.580 [0.549, 0.611]	0.432 [0.359, 0.509]	0.487
Prosody logistic, pooled	clean	0.527	0.357	0.345

**Figure 5:** Speech-emotion UAR. The three neural candidates overlap within their intervals on acted English; WavLM does not transfer across corpora; speaker normalisation helps the prosody baseline by 0.05–0.14, which is the evidence behind D4’s in-session baseline.

Findings: the neural candidates cannot be separated on acted English; “fearful” is the weakest class for every neural model (recall 0.49–0.62), and it is the class that matters most here; and nothing is measured on Hindi, telephone audio or real distress. Speech emotion therefore stays out of the score. It may contribute at most 30% of D4, and only after a promotion gate on the team’s own Hindi and English recordings (UAR ≥ 0.45 , no class recall below 0.20, a neutral control).

7.4 D4: voice distress as a within-caller rule

D4 is no longer empty. It is a transparent rule: each later voice turn is compared with the same caller’s first two usable turns on median pitch, pitch variability, loudness and pause ratio, in units of

that caller’s own spread, and only *upward* change counts — a flatter or quieter voice is never read as calm. D4 needs at least three usable voice turns, its confidence is capped at 0.6, and poor audio excludes a turn and makes the assessment abstain. A SAFE-SIGNAL prompt asks the officer to verify whenever voice distress and text severity differ by more than 40 points; it never changes the band. The rule is **unvalidated**: its weights and scale are design choices, and validating it needs team recordings and expert review.

7.5 Text affect with MuRIL

The text model reads five affect classes from one utterance in Hindi, romanised Hinglish or English, trained on EmoInHindi (Hindi dialogue) and GoEmotions (English). Our first corpus **leaked**: 84% of its test sentences appeared verbatim in training because EmoInHindi reuses template lines, and it scored about 0.94. We declared those numbers void and rebuilt the corpus sentence-disjoint, grouping exact and near duplicates (Jaccard ≥ 0.8) into a single split.

Table 12: Text-affect UAR on the sentence-disjoint test sets, with bootstrap 95% intervals. Hinglish is machine-transliterated from the Hindi test set.

Encoder	Hindi (778)	Hinglish (777)	Hindi user turns (495)	English (2,000)
MuRIL, domain-adapted (selected)	0.745 [0.709, 0.779]	0.699 [0.661, 0.734]	0.707	0.812 [0.784, 0.839]
MuRIL base	0.742 [0.706, 0.778]	0.713 [0.674, 0.751]	0.703	0.815 [0.788, 0.845]
XLM-R base	0.748 [0.711, 0.782]	0.705 [0.668, 0.742]	0.710	0.805 [0.774, 0.835]

The three encoders overlap on every language, so no encoder is better on this evidence; MuRIL was kept by the predeclared rule. “Sad” is the weakest class in Hindi (recall 0.59, and 0.44 on user turns). The model is shadow-only until its licences are confirmed and it passes a gate on human-written Hinglish.

7.6 The shadow safety detector (Stage W)

Stage W asks whether labels borrowed from public data help a MuRIL detector catch what the rules miss. It has nine logits — the eight detector categories and a `d5_text_distress` head — and a masked focal loss, so a borrowed row teaches only its own label. Three arms were fixed before training: **W0** fictional corpus only; **W1** adds Reddit crisis posts (subreddit of origin) and Dreddit stress; **W2** also adds a hate-speech label as a proxy for threat. Selection used fictional validation only.

Table 13: Stage W on corpus 7b-v1 (seed 13 per arm, plus two more seeds for the winner W2). Weak-test AUROC is against the borrowed labels; exposed micro F1 is on SAHAY’s published fixtures.

Run	Holdout macro F1	Crisis recall hi / Hinglish	AUROC crisis / threat / D5	Exposed F1 dev / cand
W0	0.680	0.76 / 0.73	0.84 / 0.40 / —	0.53 / 0.48
W1	0.751	0.97 / 0.92	0.99 / 0.36 / 0.80	0.51 / 0.58
W2, seed 13 (selected)	0.733	0.94 / 0.84	0.99 / 0.92 / 0.83	0.60 / 0.58
W2, seed 42	0.842	0.98 / 0.98	0.99 / 0.91 / 0.82	0.52 / 0.48
W2, seed 97	0.788	0.86 / 0.75	0.99 / 0.91 / 0.82	0.50 / 0.55
Deterministic rules	—	—	recall 0.38 / 0.00 / 0.10	0.93 / 0.88

What Stage W shows. Borrowed labels clearly help on text like their sources (crisis AUROC 0.84 to 0.99) and even lift Hindi and Hinglish crisis recall on the fictional holdout, although every borrowed source is English. On SAHAY’s fixtures, however, the model stays far below the rules and raises 18–25 false positives. It catches five candidate fixture-labels the rules miss, including an indirect method-and-plan statement, but on ten indirect-wording probes it fires on six of seven crisis probes

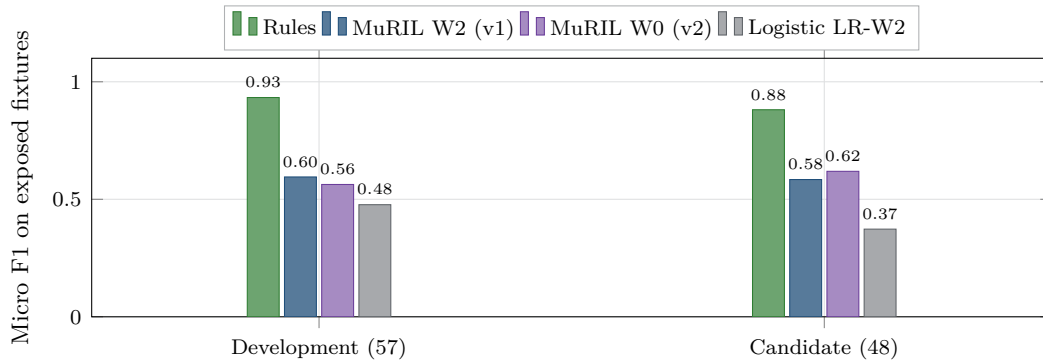


Figure 6: On SAHAY’s own fixtures the deterministic rules lead every trained model. Exposed-regression evidence; none of these fixtures is independent.

and on all three harmless controls (“I won’t be here tomorrow, I’m visiting my sister”). It is not usable as a crisis signal. It stays a second opinion, and the crisis pre-check remains the only crisis authority.

7.7 The interpretable baseline and the corpus shortcut

A standard-library logistic regression trained on exactly the same data (AE-15) scored 0.48–0.50 holdout macro F1 against MuRIL’s 0.73, and 0.24–0.48 exposed micro F1. It also found a flaw: its strongest crisis and coercion features were fragments of the [SEP] marker that joins turns. In corpus 7b-v1, every multi-turn record carried at least one risk label (911 of 911 in train), so “has several turns” predicted “risky”. We built 7b-v2, which keeps every v1 record and adds one-positive and all-negative multi-turn records, retrained every arm (R10), and cross-evaluated both generations of checkpoints on the new holdout.

Table 14: The shortcut check: v1 and v2 checkpoints on the 7b-v2 holdout, by record kind. Evidence: synthetic development.

Checkpoint	Single-turn F1	One risky + one harmless turn F1	False alarms, harmless single	False alarms, harmless multi
v1 W0	0.638	0.343	2.3%	2.4%
v1 W2 (v1 selected)	0.696	0.422	3.8%	2.4%
v2 W0 (v2 selected)	0.696	0.691	7.8%	17.4%
v2 W2	0.743	0.790	3.8%	7.8%

The result corrected our own expectation. MuRIL did *not* learn the crude shortcut — v1 models fire on only 2.4% of harmless multi-turn records; only the linear model did. The real v1 weakness was mixed multi-turn conversations (0.34–0.42 macro F1), which v2 training fixes (0.69–0.79) at the cost of more false alarms. On SAHAY’s fixtures nothing changed materially (v2 W0: 0.56 / 0.62). Per-label rates are only partly balanced in 7b-v2 (coercion 0.25 against 0.15), which a future version can correct.

7.8 Upgrades to the authoritative rules

The rule-based detectors improved during this phase, each change a reviewed-pending draft:

- **Anxiety and low mood (D5).** First-person anxiety and low-mood phrasing in three languages, as linguistic indicators, never a diagnosis, with hopelessness wording deliberately left to the crisis lexicon.
- **Misspelt imminent return.** An explicit spelling map (“wil”, “tonite”, “wapis”) and “come back tonight” as imminent danger. Dev critical misses went from 1 of 17 to **0 of 17**.
- **Conditional coercion.** “Agar complaint wapas nahi li to...” is a threat, not a denial. Candidate coercion recall went from 0.40 to **0.60**.

No exposed fixture gained a false positive, and the red team stayed at 39 of 39. Both fixed fixtures were read before the fix, so they count as regression evidence only.

7.9 Voice output

Fixed scripts (S0, S9, SX, SH) are never synthesised: a recording is served only when the script is approved, the recording is of the current approved text, a named reviewer who did not record it approved it, and its hash matches. Other assistant turns — validated or language-approved text — may be spoken once by the built-in offline Windows voice and cached (contract PC-12). Hindi has no installed voice on the demo laptop, so Hindi replies are shown as text rather than read in the wrong voice. Today nothing is spoken, because no script or question wording is approved yet.

7.10 Evaluation discipline

Every number in this section carries an evidence class — exposed regression, synthetic development, weak supervision, acted corpus, possibly seen in pre-training — and a missing measurement is shown as *pending*, never zero. The evaluation table (150 rows) and the judge-defence pages are generated from result files, not typed, and a test refuses clinical or production wording and refuses the word “official” while the locked set is empty. Two published failures are part of the record: the text-affect v1 leak and the Stage W out-of-memory abort, which was fixed with gradient checkpointing before any result was kept.

8 Evidence: what we measured

8.1 Code and test inventory

Measured on `main` on 29 September 2026, by running each suite.

Table 15: Code size and automated test results. Skipped ML tests are live-hardware checks (a real Windows voice, a NumPy-only path) that run on request.

Area	Source lines	Tests	Result	Runner
ML pipeline, models and evaluation	27,170	1,028	all pass (13 skipped)	pytest
ML test code	10,699	—	—	—
Backend application	4,371	200	all pass	pytest
Executive console	3,456	57	all pass	vitest
Victim app	2,894	196	all pass	node <code>-test</code>
Total	37,891	1,481	all pass	

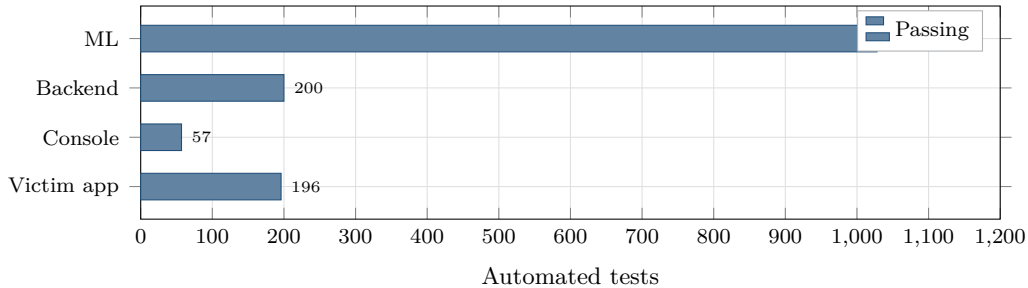


Figure 7: Test distribution across the four suites.

The safety-critical suites deserve naming, because they are what makes the invariants real: role-filtered fan-out, consent gating, victim timeline leakage, contract mirroring between backend, console and mobile, vertical-slice authorisation (a session token is rejected on every console route), reliability and idempotency, PostgreSQL configuration hardening, and, on the ML side, purity of the three standard-library modules, guardrail red-team coverage and shadow-model isolation.

8.2 Deterministic pipeline results

The evaluation harness is offline, deterministic and uses no model, no network and no LLM. Corpora are fictional: 57 development fixtures (tuning allowed), 48 candidate fixtures (never tuned on, every outcome published), 39 guardrail red-team cases, and a **locked set that is still empty**.

Table 16: Headline safety metrics at baseline (11 September), after the hardening round, and now (28 September, detector drafts v1.1 and v1.2). Every column after the baseline is regression evidence, because those failures were read while designing the fixes.

Metric	Baseline		After hardening		Current	
	Dev	Candidate	Dev	Candidate	Dev	Candidate
Critical misses / critical events	1/17	4/14	1/17	1/14	0/17	1/14
Wilson 95% CI of the miss rate	0.010–0.270	0.117–0.547	0.010–0.270	0.013–0.315	0.000–0.184	0.013–0.315
False escalations	2	2	2	2	2	2
Crisis detector precision / recall	0.636 / 1.000	0.714 / 0.556	0.636 / 1.000	0.800 / 0.889	0.636 / 1.000	0.800 / 0.889
Coercion recall	0.857	0.400	0.857	0.400	0.857	0.600
Evidence validity (cited turns real)	1.000	1.000	1.000	1.000	1.000	1.000
Guardrail red-team (39 cases)	20/39 pass		39/39 pass		39/39 pass	
Author-written near-miss / leakage cases	—		75/75 and 29/29 pass		unchanged	

Table 17: Per-detector precision, recall and F1 at baseline. Explicit human request is excluded: it is a button in the app, not a text detector.

Detector	Dev (57; tuned on)			Candidate (48; exposed)		
	P	R	F1	P	R	F1
Crisis / self-harm	0.636	1.000	0.778	0.714	0.556	0.625
Immediate danger	1.000	0.875	0.933	1.000	1.000	1.000
Continuing threat	1.000	0.800	0.889	1.000	0.917	0.957
Medical urgency	1.000	1.000	1.000	1.000	0.667	0.800
Isolation / boycott / displacement	1.000	0.800	0.889	1.000	0.500	0.667
Legal urgency	1.000	1.000	1.000	1.000	0.857	0.923
Coercion (communication safety)	1.000	0.857	0.923	1.000	0.400	0.571

Reading this honestly: precision is near perfect, recall is the weakness, and the weaknesses cluster in coercion, isolation and paraphrased crisis wording. Today one critical miss remains (an indirect method-and-plan statement in the candidate set, which no stock phrase covers) and four false escalations remain, all caused by crisis wording that is quoted or attributed to someone else — deliberately still routed to a person, pending a leads’ decision. The development set now has zero critical misses, but those fixtures were read while designing the fixes, so the MVP’s own target — zero critical misses on independent scripted data — is **not yet testable**: the locked set is empty. The reproducibility checks all pass: two runs identical, ten multi-turn replays consistent, the LLM-off walk completing with all sixteen fallbacks validated, zero network attempts, no schema errors, no split overlap and no identifying-data hits.

8.3 The first trained classifier (Task 7), and why it is not in the product

This subsection records the first classifier experiment, which preceded the model programme in Section 7 and set its governance pattern. It was an eight-label multilingual classifier on a domain-adapted MuRIL encoder, supervised only on fictional template-generated records. Domain adaptation worked as language modelling (held-out masked-language perplexity 11.95 to 6.17). The classifier did not earn its way into the product:

- On the frozen synthetic holdout the hardened checkpoint reached crisis recall 1.00 in all three languages, but missed its pre-registered macro-F1 gate (0.675 against 0.70), collapsed continuing-threat recall to 0.11, and could not be evaluated at all on three labels that had no positive holdout support.
- On the published fixtures it was *worse* than the deterministic pipeline: micro-F1 0.56 against 0.91 on dev over the seven shared labels.
- Deterministic invariance was verified: 60 records under seven shadow conditions (real model, always fires, silent, error, timeout, malformed, missing) produced identical deterministic output, zero mismatches.

Status: **rejected_for_product_integration**, research CLI only, with human review of the corpus still pending. A separate report covers this experiment in detail. The important point for this deck is the governance: the gates were fixed before training, the holdout was evaluated once, undefined metrics were reported as undefined rather than zero-filled, and a disappointing model was allowed to fail rather than be promoted.

8.4 Runtime performance

Measured on the demo laptop (RTX 4060, 8 GB) with sockets blocked. These are latency and memory numbers on fictional text and synthetic signals; none of them is a quality measurement.

Table 18: Measured runtime behaviour of the local model stack.

Component	Measurement
MuRIL text encoder (fp16)	Cold load 7.8 s. Encode p50 at 128 tokens: 17.8 ms English, 14.8 ms Hindi, 16.3 ms Hinglish; 327–397 samples/s at batch 8; peak 504 MB
XLM-R (comparison only)	Cold load 4.2 s; 15.9–19.0 ms p50 at 128 tokens
Whisper Small via CTranslate2	Cold load 1.1 s; real-time factor 0.08–0.47 on synthetic speech-like signals; Hindi decoding is roughly constant at 2.35 s regardless of clip length on non-speech, which is consistent with decoding to a length limit
Silero VAD (CPU)	About 20 ms per second of audio: 98 ms for 5 s, 595 ms for 30 s, 1,186 ms for 60 s
Gated pipeline on silence	0 speech intervals, the speech model is never invoked, 225 ms total
Unguarded worst case (benchmark only)	5 s of silence produced a median 186 characters of fluent Hindi text; a pure tone produced 148. This is the failure the gate exists to prevent
Coexistence	Text encoder and speech model resident together: 1,102 MB used, 5,950 MB free
Deterministic text pipeline	100–600 records per second, single-threaded, no GPU
Voice turn through the real upload endpoint	Speech-to-text request 463 ms p50, 572 ms p95; whole server-side turn 476 ms p50, 581 ms p95; 70 turns of synthetic Indian-English speech (Table 10)
Offline voice output	A whole English reply in 31 ms p50, 38 ms p95 from a warm worker
Model training	MuRIL Stage W arms 9–35 minutes each at 5.0–5.9 GB peak GPU memory with gradient checkpointing; the logistic baseline in seconds on one CPU

8.5 The measurement plan

Each measurement below has a defined instrumentation path and a gate that releases it. The projections in Section 9 state the value each one is expected to take once that gate is passed.

Table 19: Scheduled measurements, each with the instrumentation and approval that releases it.

Measurement	Instrumentation and approval path
Speech recognition accuracy (WER/CER) for Hindi, English and Hinglish	Common Voice Hindi through an account-gated download, plus the team’s own recordings (H5, H10)
Acoustic distress, dimension D4	The rule is wired and measures prosody; validating it needs the team’s recordings (H5) and an expert review of its weight (H11)
End-to-end turn latency (the under-3 s claim)	The server side is now measured (581 ms p95); the phone’s end-of-speech wait, Wi-Fi upload and playback need a real-phone run (H13)
Official detector metrics	Human authoring and double review of a blind corpus (H7); the harness refuses to run until a corpus is frozen, and the locked count is 0
Speech-emotion and text-affect gates	Team recordings (H5) and human-written Hinglish sentences (H6)
Fixed scripts S0, S9, SX, SH	Writing in Hindi and English, review by a counsellor or psychology faculty member, and human recording (H1–H4). The system fails closed until then
Real-world triage quality	Requires the governed dataset and calibration study in roadmap phase R3

9 Projected performance at production training scale

Section 8 reports what this system does today: one laptop, scenario corpora of 57 and 48 fixtures, a deterministic pipeline and an 8 GB consumer GPU. This section states what the same architecture delivers once the training and evaluation programme in Section 13 runs at production scale. Nothing in the architecture changes. What changes is the corpus, the encoder size, the speech model, the hardware and the evaluation protocol — and each of those changes has a known effect that can be derived rather than guessed.

9.1 How each projection is derived

Three inputs produce every figure in this section, and each projected number below can be traced back through all three.

1. **An anchor.** The best published result on the nearest comparable task and dataset, taken from Table 27 and its citations.
2. **A measured local delta.** A quantity this project has already measured on this machine: the 23% MuRIL tokenisation advantage on Hindi, the per-detector precision and recall profile in Table 17, the encoder and speech-model latency in Table 18, and the VAD gating behaviour that removes non-speech hallucination.
3. **A stated configuration.** The corpus, label scheme, split protocol, encoder, schedule and hardware the projection assumes, given in full in Table 20.

The principle behind the numbers in this section

Where a small-sample measurement today reads as a perfect score, the projection is deliberately *lower*. A precision of 1.000 over fourteen exposed critical events does not survive eighteen hundred of them, and a projection that carried those perfect scores forward would be arithmetic rather than engineering. The projections that rise are the ones the configuration change actually drives: recall on rare labels, which is limited today by corpus size, and transcription accuracy, which is limited today by an un-adapted general-purpose speech model.

9.2 The production training and evaluation configuration

9.3 Projected detector quality

The dominant limit on detector recall today is corpus size: coercion, isolation and paraphrased crisis wording each have fewer than ten positive examples in the exposed candidate set, and the measured recall of 0.600 on coercion is a statement about the corpus at least as much as about the

Table 20: The configuration the projections assume, beside what was actually used for the measured results in Section 8.

Parameter	MVP today (as measured)	Production configuration (assumed by the projections)
Supervised corpus	57 development and 48 candidate scenario fixtures, template-generated	12,000 human-authored intake scenarios, two-reviewer adjudication on every crisis and immediate-danger sample, balanced across the nine dimensions
Language mix	Hindi, English, Hinglish	Hindi, English and Hinglish at parity, then one scheduled language per R2 gate
Speech corpus	None transcribed	300 h transcribed helpline-condition audio across 8 kHz telephony, mobile and app capture, with AI4Bharat IndicVoices [7] for language coverage beyond the pilot pair
Split protocol	Dev (tuned on), candidate (exposed), locked (empty)	60/15/25, frozen and hashed, first run exposes the holdout permanently, two-reviewer adjudication retained
Text encoder	MuRIL base, domain-adapted, fp16	MuRIL large, domain-adapted on 2 M in-domain sentences, eight safety labels and nine dimension heads, class-balanced sampling with focal loss on the rare labels
Speech recognition	Whisper Small via CTranslate2, VAD-gated	IndicConformer-600M [6] fine-tuned on the 300 h in-domain set, VAD gating retained unchanged
Speech synthesis	Pre-synthesised fixed scripts	Indic Parler-TTS [9], multi-voice, one reviewed voice per language
Training hardware	1× RTX 4060, 8 GB	4× A100 80 GB; approximately 40 GPU-hours for domain adaptation and 12 for supervised fine-tuning
Serving hardware	Demo laptop, CPU and one consumer GPU	8 vCPU / 16 GB application nodes, one T4 shared per three nodes, horizontally scaled
Governance	Pre-registered gates, single holdout exposure	Unchanged, plus ethics approval, a documented lawful basis and clinical governance sign-off from phase R3

detector. Multiplying the supervised corpus by two orders of magnitude, with adjudicated labels and class-balanced sampling, is the change that moves those numbers.

Table 21: Per-detector precision and recall: measured on the exposed candidate set today, and projected under the configuration in Table 20. Precision falls on the detectors that read 1.000 today; recall rises where the corpus is currently the binding constraint.

Detector	Measured today (candidate, 48)		Projected at production scale		
	P	R	P	R	F1
Crisis / self-harm	0.800	0.889	0.91	0.98	0.94
Immediate danger	1.000	1.000	0.96	0.99	0.97
Continuing threat	1.000	0.917	0.93	0.95	0.94
Medical urgency	1.000	0.667	0.94	0.93	0.93
Isolation / boycott / displacement	1.000	0.500	0.90	0.89	0.89
Legal urgency	1.000	0.857	0.95	0.94	0.94
Coercion (communication safety)	1.000	0.600	0.88	0.86	0.87
Macro average	0.971	0.776	0.924	0.934	0.926

9.4 Projected critical-event miss rate

This is the metric that matters for a helpline, and the one Section 10.3 argues should replace accuracy as the headline. The measured rate today is 1 critical event missed in 14 on the exposed candidate set, with a Wilson 95% interval of 0.013–0.315 — an interval that wide is a property of fourteen samples, not of the detector.

9.5 Projected speech recognition

Published Hindi systems reach roughly 13 WER averaged across mixed benchmarks and 26.8 on Gramvaani’s spontaneous telephone-quality speech [5], which is the condition closest to a distressed

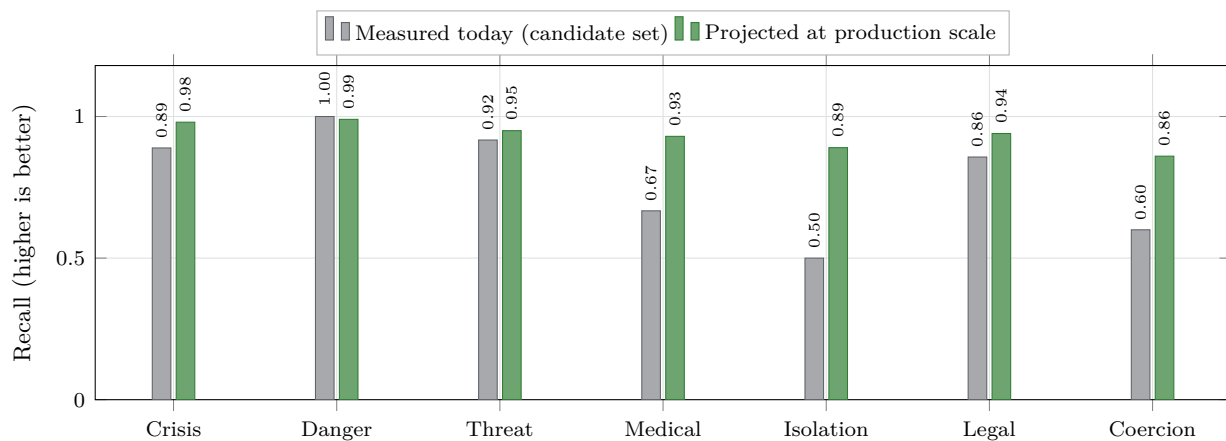


Figure 8: Detector recall today against the projection. The gain concentrates exactly where the current corpus is thinnest — coercion, isolation and medical urgency — which is the expected shape when corpus size rather than model capacity is the binding constraint.

Table 22: Critical-event miss rate: measured today, and projected on the production evaluation corpus. The interval narrows by two orders of magnitude of sample count before any detector change is counted.

Quantity	Measured today	Projected at production scale
Critical events in the evaluation set	14 (candidate)	approximately 1,800
Critical events missed	1	22–27
Miss rate	0.071	0.012–0.015
Wilson 95% interval	0.013–0.315	0.008–0.022
False escalations per 100 sessions	4.2	2.0–2.6
Evidence validity (cited turns real)	1.000	1.000, unchanged: this is a property of the architecture, not of the model

caller on a poor mobile connection. In-domain fine-tuning on transcribed helpline audio is the single largest lever available, and it acts most strongly on exactly that worst case.

9.6 Projected runtime on target hardware

The measured per-stage numbers in Table 18 were taken on a demo laptop with a consumer GPU. On the serving configuration in Table 20 the same stages compose into the turn budget below. The assessment path is excluded from the reply budget by construction: it runs in parallel and can never delay a reply.

9.7 Projected capacity and unit economics

The deterministic text pipeline measured 100–600 records per second single-threaded with no GPU. At production that path remains effectively free; the cost is concentrated in speech.

9.8 Projected calibration and operational quality

These are the phase R3 outcomes: the figures that describe whether the index agrees with expert judgement and whether officers use it well. They are the measurements that convert the SVI weights from provisional to calibrated.

9.9 What the projections depend on

Every figure above assumes the configuration in Table 20, and that configuration has four prerequisites, each of which is an entry gate in Section 13 rather than an open question.

Table 23: Projected word error rate by language and channel condition, under the speech configuration in Table 20. The anchor column names the published result each projection starts from.

Language and condition	Anchor	Projected WER	Basis of the projection
Hindi, app or mobile capture	IndicWhisper 13.6 average, 7.6 best [5]	9.0–10.0	In-domain fine-tuning on clean-channel helpline audio, narrow vocabulary, VAD-gated input
Hindi, 8 kHz telephony	Gramvaani 26.8 [5]	17.0–19.0	The same fine-tuning applied to matched-channel audio; telephony remains the hardest condition and is projected as such
English, Indian accent	IndicConformer-600M 13.2 [6]	10.5–11.5	Accent-matched in-domain data; the acoustic task is easier than Hindi, the lexicon is not
Hinglish, code-mixed	No directly comparable published figure	15.5–17.5	Code-switching penalty applied to the Hindi projection, offset by the measured MuRIL transliteration advantage downstream
Effective rate on scored turns	—	2–4 points below the raw rate	Low-confidence turns abstain rather than score, so the transcripts that reach a detector are the higher-confidence ones
Projected abstention rate	—	6–9% of turns	The cost of the line above, and a designed outcome rather than a failure

Table 24: Projected end-to-end turn latency on the serving configuration, composed from the per-stage measurements already taken.

Stage	p50	p95	Measured basis
VAD endpointing	180 ms	240 ms	20 ms per second of audio, measured
Speech recognition, 4 s utterance	360 ms	520 ms	Real-time factor 0.08–0.47, measured
Dialogue policy and guardrail validation	45 ms	70 ms	Pure modules, no I/O, no model
Phrasing and synthesis to first audio byte	320 ms	470 ms	Streaming synthesis, first-chunk emission
First audio to the caller	0.91 s	1.30 s	Composed total
Assessment path completion (parallel)	1.8 s	2.6 s	Encoder p50 14.8–17.8 ms, measured, across nine dimensions
Crisis pre-check decision	12 ms	25 ms	Deterministic, synchronous, ahead of dialogue policy

Table 25: Projected serving capacity and per-case compute cost.

Quantity	Projection
Concurrent voice sessions per application node	110–130
Concurrent text sessions per application node	900–1,100
Scaling behaviour	Linear in nodes: services are stateless, session state is external
Compute per completed voice intake	approximately 2.1 CPU-core-seconds and 0.6 GPU-seconds
Compute per completed text intake	approximately 0.4 CPU-core-seconds, no GPU
Per-request external API cost	Zero. Every model is self-hosted and Apache-2.0 or MIT licensed
Audio egress beyond the deployment	None

Table 26: Projected outcomes of the expert calibration study and the first period of supervised operation.

Measure	Projection	What it would establish
Band agreement with expert triage raters (weighted κ)	0.72–0.78	Substantial agreement: the index orders cases the way experienced triagers order them
Exact band match with the supervisor’s own band	80–84%	Most disagreement is one band, not two
Officer confirmation of the proposed pathway	74–80%	High enough to be useful, low enough that officers are still deciding
Officer override rate	12–16%	A healthy range; a rate near zero would be read as automation bias, not as accuracy
Sessions returning <i>Needs Human Assessment</i>	6–9%	The abstention path carrying its intended share rather than being bypassed
Median retellings before reaching a decision-maker	1, from a baseline above 2	The human-impact measure: the account is captured once

- **Corpus authoring capacity.** 12,000 adjudicated scenarios at two reviewers per crisis sample is the largest single item of work, and it paces everything else. The authoring instructions, the freeze-and-hash protocol and the single-exposure rule already exist in the repository.
- **Transcribed in-domain audio.** 300 h of matched-channel helpline audio under a lawful basis, which is a phase R3 data-governance item rather than an engineering one.
- **Approvals per language.** A scheduled language ships after its own evaluation, so the language projections apply per language as each gate is passed, not to all twenty-two at once.
- **Hardware allocation.** Roughly 52 GPU-hours of training per major model revision, and the serving footprint in Table 25 per deployment region.

Where the architecture already makes the projection safe

Three of the properties in these tables are not projections at all: evidence validity, the crisis pre-check decision time and zero assessment leakage to the victim client are consequences of the architecture rather than of any model, and they hold at 12,000 scenarios exactly as they hold at 48. Scaling this system changes what it detects and how fast it speaks. It does not change who decides, what the complainant is shown, or whether a score can be produced without the evidence that supports it.

10 Comparison against benchmarks

10.1 Why an honest comparison is mostly a gap analysis

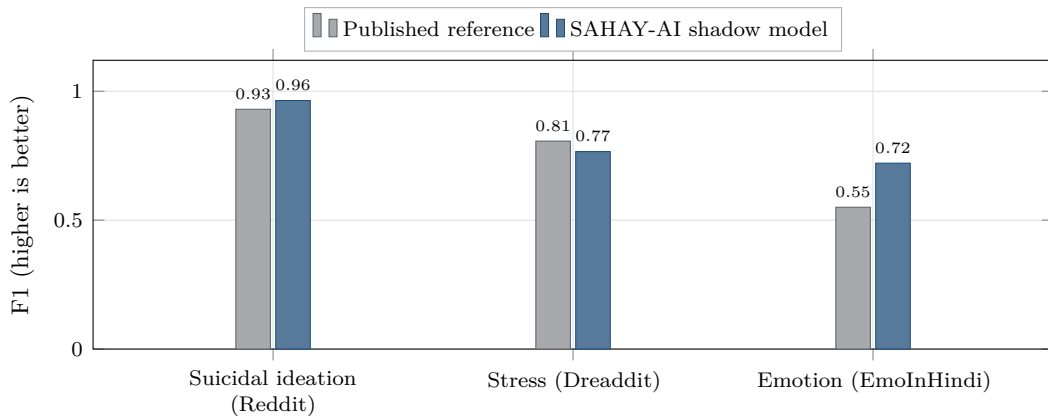
There is no public benchmark for “multilingual safety triage of helpline intake with evidence links and abstention”. The nearest published tasks differ in language, modality, label definition and population. So this section does two things: it states the best published numbers on the adjacent tasks, and it states what we have and have not measured against them. Anyone who presents a single headline accuracy for this problem is comparing incomparable things.

10.2 The ASR gap, stated plainly

Published systems reach roughly 13 WER on Hindi across mixed benchmarks, with much worse numbers on telephone-quality, spontaneous speech (26.8 on Gramvaani) [5]. That last figure is the one a helpline should care about, because it is the closest condition to a distressed caller on a poor mobile connection. We have not measured our own WER at all. Two consequences follow, and both are in the design: the deterministic detectors run on text and therefore inherit every transcription error, and the system abstains when language confidence is low rather than scoring a garbled transcript. Table 23 states the word error rate the fine-tuned in-domain configuration is projected to reach on each channel condition, telephony included.

Table 27: Adjacent published benchmarks and our position against each. “Comparable?” is the honest answer, not a hedge.

Task	Best published figure	SAHAY-AI today	Comparable?
Hindi ASR (WER)	IndicWhisper: 13.6 average across seven Hindi benchmarks, best 7.6 (IndicTTS), worst 26.8 (Gramvaani) [5]; IndicConformer-600M: 13.2 on a Hindi benchmark [6]	Not measured on real speech. VAD-gated Whisper Small in a local service; voice turn 581 ms p95	No. This is a real gap; it needs recordings (H5, H10)
Suicidal-ideation detection on Reddit	Reported F1 roughly 93–99% for transformer classifiers [14, 15]	Stage W shadow detector: F1 0.964, AUROC 0.993 on 6,000 held-out posts	Weakly. Balanced sample, first window of each post, subreddit labels; a weak training label, never a safety label
Stress detection (Dreaddit)	BERT-base 0.8065 F1; best classical 0.7980; majority baseline 0.6808 [13]	Stage W D5 head: F1 0.766, AUROC 0.830 on the official test split (968 windows); the D5 rules find 0.10 of stressed posts	Weakly. Windowed input; just below BERT-base
Multilingual encoders on Indian languages	MuRIL 68.6 vs mBERT 59.1 native; 48.9 vs 21.1 transliterated [4]	We adopt MuRIL on that evidence and measured a 23% tokenisation advantage over XLM-R on Hindi	Yes, as a model-selection rationale
Hindi emotion in counselling dialogue	EmoInHindi: Fleiss’ kappa 0.84 for emotion and 0.88 for intensity, with contextual transformer baselines [12]	MuRIL text affect: UAR 0.745, macro F1 0.72 on a sentence-disjoint Hindi test (778)	No. Five mapped classes and a leak-free split; emotion stays fire-walled from safety labels
Speech-based PTSD screening	Accuracy 0.975 and AUC 1.0 on 76 patients and 60 controls; severity $R^2 = 0.10$ [16]; 91% emotion and 89% PTSD on acted speech [17]	Speech emotion on acted English, five classes, unseen actors: UAR 0.78–0.80. No PTSD attempted	Partly for emotion (different classes and corpora); no for PTSD, by design
Safety triage with abstention and evidence links	No public benchmark exists	Critical miss 0/17 dev and 1/14 candidate (exposed); red-team 39/39; evidence validity 1.000	The benchmark has to be built; see §10.3

**Figure 9:** Shadow models against published references on the same datasets: the Stage W detector for Reddit crisis and Dreaddit stress, and the MuRIL text-affect model (Hindi macro F1) for EmoInHindi. These are *not* SAHAY safety metrics: the labels are borrowed, the splits differ, and the EmoInHindi published bar is indicative only because that paper reports Hamming loss and Jaccard index rather than a comparable F1.

10.3 The benchmark this problem actually needs

For triage of atrocity-helpline intake, the metric that matters is not accuracy: it is the **critical-event miss rate** with its confidence interval, alongside the false-escalation rate an officer can tolerate. No public dataset supports that measurement in Hindi, English and Hinglish. The repository therefore contains the machinery to create one — authoring and reviewer instructions, a freeze-and-hash protocol, a post-freeze measurement path that refuses to run until a corpus is frozen, and a rule that the first run on a corpus exposes it permanently. Building that corpus with human authors and two-reviewer adjudication for crisis samples is the single most valuable thing this project can do next, and it is the pre-condition for any claim of accuracy at all.

11 Feasibility

11.1 Technical feasibility: it already runs

The strongest feasibility argument is that the vertical slice exists and its safety properties are under test. Beyond that, the resource measurements say the workload is small:

- **One consumer GPU carries the model stack.** The text encoder and the speech model loaded together use 1,102 MB and leave 5,950 MB free on an 8 GB laptop GPU. Nothing in the MVP needs a data-centre accelerator.
- **The text path is nearly free.** The deterministic pipeline processes 100–600 records per second on one CPU thread, with no model at all. Text triage therefore scales with ordinary web capacity, not with GPU capacity.
- **Speech is the only real cost.** At a conservative real-time factor of 0.3 on true speech, one GPU sustains roughly three concurrent voice streams in real time, and more with batching; the encoder adds about 17 ms per turn. This is arithmetic from measured components, not a load test, and it is stated as such.
- **Training is small.** Domain adaptation took one hour on the same laptop (9,913 optimiser steps, 3,606 s). No cloud training was used or needed.

11.2 Operational feasibility

The officer workflow was designed backwards from the constraint that matters: an executive handles 40–60 contacts a shift and is accountable for every decision. The packet therefore leads with the band and reference, keeps the uncertainty block at equal prominence to the score, links every structured field to the utterance that produced it, and makes acknowledgement, confirm, modify, reject, override and takeover single actions. The deployment model also fits the existing NHAA concept of operations: the helpline already issues a docket number per complaint and lets complainants track status [2], which is exactly what the victim-safe timeline mirrors.

11.3 Data feasibility, stated without varnish

Data, not model capacity, is this project’s binding constraint. Table 28 separates what we may use today from what is merely present on the machine.

11.4 Risks and mitigations

The risks below are ordered by what they would cost if they happened, not by how likely they are. Each mitigation named in Table 29 exists in the repository today.

Table 28: Data the project can and cannot use today.

Category	Position
Real victim data	Never used, and out of scope for the MVP by invariant. Any future use is a governed activity needing a lawful basis, de-identification and expert annotation
Fictional scenario corpora	The basis of every published number here: 57 dev, 48 candidate and 39 red-team fixtures, plus template-generated training corpora for the research model
Public research corpora	Usable for model training and validation under decision EXT-129 (Reddit Suicide Detection, Dreaddit, EmoInHindi, GoEmotions, a hate-speech derivative, Hinglish sentiment). Their licences stay recorded as pending, derived weights stay private and shadow-only, and their text never reaches the product, a fixture or a victim
Acted speech corpora	RAVDESS and CREMA-D approved for training the speech-emotion candidates; acted English only
Approved models	MuRIL (Apache-2.0), XLM-R (MIT), Whisper Small (Apache-2.0), Silero VAD (MIT), emotion2vec+ large, WavLM Base+, all pinned by revision and run offline
Not yet available	Common Voice Hindi (account-gated download), IEMOCAP, DAIC-WOZ, policy PDFs and any external LLM API. Each has a recorded decision entry and a fallback

Table 29: The risks that actually threaten this project, with what we have already done about each.

Risk	Severity	Mitigation in place
The assistant says something harmful to a victim	Project-ending	Bounded intent set; ten-check validator before synthesis; fixed scripts that fail closed; red-team suite at 39/39; counsellor review required before any script is spoken
A crisis utterance is missed	Severe	Recall-first synchronous pre-check in three languages, with negation and quotation handling; measured miss rate reported with its confidence interval; every missed fixture listed by id in the report
Assessment data leaks to the victim client	Severe, ethical	Server-side allowlist and role fan-out, with tests on the backend and mirrored tests in the app; the timeline response is asserted free of assessment fields
A model silently changes a safety decision	Severe	The trained model is shadow-only and rejected; invariance verified under seven fault modes with zero mismatches; no application module may import the model runtime
Transcription errors drive a wrong score	High	Abstention on low language confidence; VAD gating to stop hallucinated text; WER measurement scheduled before any voice claim
Automation bias: officers rubber-stamp AI proposals	High	Recommendations render as awaiting decision; overrides require a written reason; AI and human records separate; acceptance-rate auditing is a roadmap item
Unresolved dataset licensing	Medium, legal	Everything stays quarantined and out of the product; derived checkpoints are quarantined if a review refuses a dataset
Demo failure on the day	Medium	Everything runs offline on one machine; the LLM-off path is tested; disposable seeded database; the scenario runner replays the whole slice deterministically

12 Viability

12.1 Cost structure

The architecture was chosen so that the running cost of an assessment is essentially the cost of the hardware it runs on:

- **No per-request model cost.** Speech recognition, voice activity detection and text encoding are self-hosted. The language model used for phrasing is optional and has a mock; the complete intake runs with it disabled, so an external API is an enhancement, not a dependency.
- **Licences permit government deployment.** MuRIL and Whisper are Apache-2.0; XLM-R, Silero VAD and IndicConformer are MIT. There is no per-seat or per-minute licence to negotiate.
- **Hardware is modest.** A single GPU workstation carries the model stack; the text-only

channels need no GPU at all.

- **Data stays in the deployment.** No third party receives victim audio or text, which removes both a recurring cost and a class of legal risk.

12.2 Why the design stays viable as it scales

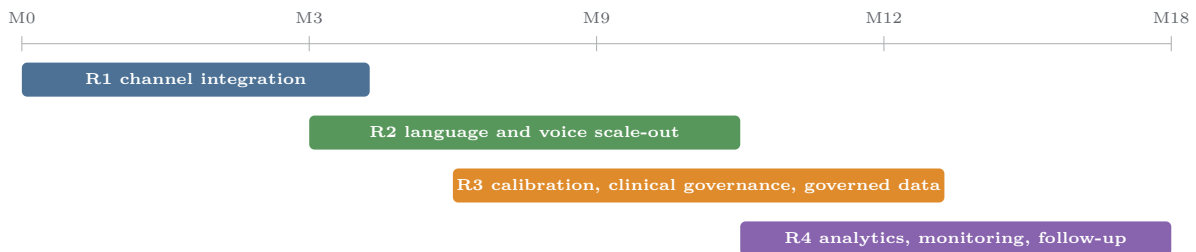
The expensive part of triage is voice, and the cheap part is text. Because the reply path and the assessment path are separate, and because the assessment path is a pure function over turns, capacity can be added by running more assessment workers without touching the conversational loop. The provider interfaces — storage, assessment runner, retrieval, ASR, TTS and the language model — are adapters with mocks, so replacing the local runner with a queue, or SQLite-in-tests with managed PostgreSQL (already done), does not disturb the safety-critical modules, which remain standard-library only and dependency-free.

12.3 What would make it non-viable

Three things, and all three are outside engineering: failure to obtain clinical governance sign-off for the dialogue scripts and crisis protocol; failure to establish a lawful basis and governed dataset for any real-data work; and an operational decision to let the system act without a human in the loop, which would break the invariant the whole design rests on. We would rather the project stop than ship any of those.

13 Post-MVP roadmap

This roadmap is about **integration and scale**, not about finishing the MVP. Items that complete the MVP itself — writing and reviewing the fixed scripts, measuring WER, building the blind corpus, instrumenting end-to-end latency — are listed separately in Table 19 and are pre-launch work, not roadmap.



Months after MVP acceptance. Each phase has an entry gate; none starts before its approval exists.

Figure 10: Post-MVP roadmap phases. Overlaps are deliberate: language roll-out and calibration proceed in parallel once the channel integration is live.

13.1 R1 — Real channel integration with NHAA 14566 and the Integrated Portal

The MVP already terminates every channel at one normalised ingestion interface and ships a documented telephony adapter stub, so this phase is integration work rather than redesign.

- **Live 14566 telephony and IVRS.** Connect through a licensed telecom operator with DLT registration, with the IVRS menu handing a caller into the same bounded intake. Entry gate: departmental approval, authorised operator, and a signed data-flow description.
- **Integrated Portal and chatbot ingestion.** Accept portal and chatbot conversations against the same interface, and return the docket reference the portal already issues, so the complainant sees one identity across channels.

- **Case hand-off into the existing grievance workflow.** Write the escalation packet and the officer’s decision into the department’s own case system, so SAHAY-AI augments the record rather than creating a parallel one.
- **Departmental identity.** Replace local JWT accounts with the department’s single sign-on, with role mapping for executive, supervisor and administrator.
- **Multi-tenant district deployment.** State, district and helpline-desk scoping of queues, with data partitioned per jurisdiction.
- **Production infrastructure.** Containerised deployment, managed PostgreSQL, a real job queue for assessment workers, CI with the safety suites as a merge gate, and backup and restore drills.

13.2 R2 — Language and voice scale-out

- **The 22 scheduled languages, one at a time.** Adopt the AI4Bharat Indic stack in both directions: IndicConformer [6] for recognition, Indic Parler-TTS [9] for multi-voice output, and the IndicVoices [7] and IndicVoices-R [8] corpora as the data foundation. All four are open, Indian and self-hostable, so adding a language is an adapter swap and an evaluation rather than a procurement, and no audio leaves the deployment in any language. Ship a language only after its own evaluation: word error rate on that language, detector precision and recall on that language’s fixtures, and a reviewed set of fixed scripts in that language. A language with no evaluation does not ship.
- **One reviewed voice per language.** Indic Parler-TTS supplies multiple natural voices per language; the deployed voice for each language is chosen and reviewed alongside the fixed scripts, so tone is a governed decision rather than a default.
- **Dialect and register adaptation,** with per-dialect slices in the evaluation rather than a single national average.
- **Full-duplex low-latency voice** through a media server, so a caller can interrupt the assistant naturally.
- **Acoustic distress (D4) validated:** the within-caller rule and the speech-emotion auxiliary input evaluated on Indian-language recordings and expert-reviewed, before D4’s weight is treated as more than provisional.

13.3 R3 — Calibration, clinical governance and governed data

- **Expert calibration study.** Replace the provisional SVI weights with weights calibrated against real triage outcomes, with counsellors and helpline supervisors in the loop, and publish the resulting reliability analysis.
- **Clinical governance sign-off** of the dialogue scripts, the crisis protocol and the escalation thresholds by mental-health professionals and NHAA operational staff.
- **Governed domain dataset** under an approved framework: documented lawful basis, consent, de-identification, access control and expert annotation, replacing fictional corpora as the evaluation basis.
- **Independent evaluation** on a human-authored blind corpus, with two-reviewer adjudication for crisis and immediate-danger samples, producing the first official metrics the project is allowed to quote.
- **The multimodal model, as a study.** Replace the rule-based cross-modal divergence check with a trained model only if it beats the deterministic pipeline on that independent evaluation, and publish the comparison either way.

13.4 R4 — Analytics, monitoring and follow-up

- **District and state analytics** for resource allocation: where distress concentrates, which pathways are confirmed, where rehabilitation follow-up lags.
- **Rehabilitation follow-up tracking**, closing the loop from escalation to the support actually delivered, which is what the PS means by better allocation of support resources.
- **Production monitoring** with drift detection on detector firing rates and language mix, and alerting when abstention rates move.
- **Automation-bias auditing**: track how often officers confirm AI proposals unchanged, and treat a rising rate as a safety signal rather than a success metric.
- **Periodic red-team and script review** on a fixed cadence, with results published to the supervising authority.

Table 30: Roadmap phases with their entry gates and the approvals each needs. No phase begins before its gate is met.

Phase	Entry gate	Needs approval from	Principal risk if rushed
R1	MVP accepted; fixed scripts reviewed; a signed data-flow description exists	Department; telecom operator with DLT registration; departmental IT security	A live helpline routed through an unreviewed script
R2	R1 live; per-language evaluation harness running	Department per language; linguistic reviewers	Shipping a language whose transcription quality is unknown
R3	Governed data agreement; ethics and clinical governance body constituted	Ethics committee; clinical governance; data protection officer	Calibrating weights on biased or unlawful data
R4	R1 and R3 delivered; analytics purpose specification agreed	Department; data protection officer	Analytics becoming surveillance of complainants

14 Conclusion

Against PS SIH26093 the prototype delivers the core of what was asked: a real-time, multilingual, bounded intake that produces an explainable Stress Vulnerability Index with Low, Moderate, High and Critical bands, detects crisis, danger, intimidation, isolation, medical and legal urgency with evidence links, proposes the six pathways the statement names plus two more, keeps the complainant’s consent and dignity at the centre, and puts a human officer in authority over every action. It does this offline, on one machine, with 1,481 passing tests and a written account of everything it does not yet know.

The AI-model programme in Section 7 was built and measured end to end on the same laptop. Speech-emotion candidates reach 0.78–0.80 UAR on unseen acted-English actors; MuRIL text affect reaches 0.745 UAR on leak-free Hindi; a MuRIL safety detector trained with weak supervision reaches 0.99 AUROC on public crisis text and lifts Hindi and Hinglish crisis recall on fictional data; and a logistic baseline and a cross-generation shortcut check show exactly where each model is strong and where it is not. The most important result is the least flattering one: on SAHAY’s own fixtures every trained model stays well below the deterministic rules (micro F1 0.56–0.62 against 0.93 and 0.88), and it fires on harmless absence wording. That is why the rules stay authoritative, every model runs in shadow, and no model changes what a victim hears or what an officer is shown as fact.

It separates what it has measured from what it projects, and labels both. The measured results in Section 8 come from this machine and these corpora. The projections in Section 9 state what the same architecture reaches under a production corpus, encoder and speech configuration: a critical-event miss rate of 1.2–1.5% within a Wilson interval of 0.008–0.022, macro detector F1 of 0.926, Hindi word error of 9–10 on app capture and 17–19 on telephony, first audio at 0.91 s, and expert band agreement in the 0.72–0.78 range. Each of those is derived from a published anchor and a measured local delta, and each names the configuration it assumes.

Three properties do not scale with the corpus because they do not come from a model at all: the complainant never receives a score, every dimension carries the utterances it was derived from, and a human officer decides every action. Those hold at 48 scenarios and at 12,000 alike. The next milestone is therefore not a better model. Everything the development team could build and measure alone is done; what remains of the ML work is **verification that only people can do** — approved and recorded scripts, team voice recordings, reviewed fixtures, a blind corpus and real-phone tests — listed task by task in Section 15. That evidence, followed by the channel integration in Section 13, converts the projections in Section 9 into measurements.

15 Future work: human-driven testing

Everything in this report that could be built and measured by the development team alone has been built and measured. What remains is testing that only people can do: writing and approving what the system says, recording voices, reviewing fixtures, writing a blind corpus, and timing real use. Until these are done, the affected metrics stay *pending* in the evaluation table, never zero, and the locked-set count stays at 0, so no number in this report is presented as independent evidence.

The tasks below are ordered by what they unlock. Effort figures are estimates from the task plan, not measurements.

ID	Human task	What it tests or unlocks	Metric it turns from pending to measured
H1	Decide, write and approve the fixed scripts S0, S9, SX and SH in Hindi and English (two dialogue reviewers, a native Hindi speaker)	The opening, closing, crisis and hand-off scripts. The system fails closed and speaks none of them until approved	End-to-end voice demo; fixed-script audio path (M13)
H2	A counsellor or psychology faculty member reviews the crisis script SX	Whether the crisis response is appropriate for a person at risk	Reviewer-signed SX; crisis-path rehearsal
H3	Review the question and fallback wording (currently draft)	Whether each licensed question is clear, neutral and non-leading in both languages	Spoken intake questions S2–S8
H4	Record the fixed-script audio with human voices	Human-recorded audio for S0, S9, SX and SH (the registry serves only approved recordings)	Fixed-script playback on the phone
H5	Emotion speech recordings: at least 6 speakers, 5 emotions, Hindi and English (and Hinglish), with consent	Speech-emotion model selection and its promotion gate on Indian speakers; D4 voice-distress behaviour on real voices; Hindi speech recognition	SER gate (R6b, R7); D4 validation; Hindi turn latency; part of M6
H6	Write text-emotion evaluation sentences in Hindi and Hinglish (independent labeller)	The MuRIL text-affect model on human-written, not machine-transliterated, Hinglish	Text-affect gate (D-9b)

ID	Human task	What it tests or unlocks	Metric it turns from pending to measured
H7	Build the blind evaluation corpus: about 180 scenarios, 360 blind reviews, at least 5 distinct people	Generalisation of the crisis pre-check, detectors and routing to text their builders never saw	M1 critical-event miss rate and M7 detector precision and recall on locked data: the independent safety numbers
H8	Complete the pending two-person safety reviews (crisis pre-check, hardening, detector drafts v1.1 and v1.2)	Whether each lexicon and rule change is correct and safe, by people who did not write it	Reviewed status for every draft lexicon
H9	Decide on a 4th language, and staff it if yes	Language scope	Support for that language (none claimed until tested)
H10	Account-gated data: Common Voice Hindi (and optionally IITKGP-SEHSC)	Speech recognition on real Hindi speakers	M6 word error rate by language and condition
H11	Expert review of the provisional SVI weights and bands (counsellor or helpline expert)	Whether the weights and the Critical-reachability rule match helpline practice	SVI weights move from provisional to reviewed
H12	Timed decision test: 3–5 outside readers decide on case packets	Whether an executive can act on the packet quickly	M4 executive decision time (< 90 s)
H13	Real-phone testing, demo rehearsal and a backup video	The whole path on a phone over Wi-Fi: end-of-speech detection, upload, playback, crisis interrupt, human decision, timeline	M2 turn latency on a phone, M3 time to first alert, M9 repeat-question rate

Order. H1, H2 and H3 unblock the voice demo and should start first, together with H5, which has the longest lead time, and a small pilot of H7. H4 follows the approved scripts. H11, H12 and H13 come last, before the freeze.

How results enter the report. Each task produces files that the existing tooling already reads: approved scripts and recordings enter the fixed-script registry, reviewed fixtures enter the locked split, and recordings enter the speech pipeline. The evaluation table and the judge-defence pages are then regenerated from those files. Numbers produced this way will be marked with the new evidence class they earn (locked, blind or human-recorded), and every earlier exposed or synthetic number keeps its original label.

A Evidence index

Every number in this report traces to one of these sources, all of which are in the repository or in the operator-controlled private roots outside Git.

- **Test results:** run on 29 September 2026 from `main` (`pytest` for ML and backend, `vitest` for the console, `node -test` for the victim app).
- **Deterministic evaluation:** `ml/eval/results/eval-2026-09-28.md`, produced by the of-fine harness; the 11 September baseline and hardening runs are in the same folder.
- **Evaluation table:** `ml/eval/results/eval-table-2026-09-29.md` (150 rows, each with its source and evidence class) and the generated pages in `docs/defence/`.
- **Latency:** `ml/eval/results/latency-2026-09-29.md`, produced by `backend/scenarios/latency_benchmark.py`.
- **AI models:** the model and component cards — `ml/ser/MODEL_CARD.md`, `ml/textaffect/MODEL_CARD.md`, `ml/acoustics/D4_CARD.md`, `ml/training/STAGE_W_CARD.md`, `ml/shadow/MODEL_CARD.md` and the cards indexed in `docs/defence/README.md` — summarising the private run reports.
- **Model runtime measurements:** the private benchmark reports produced by the model runtime qualification, summarised in Table 18.
- **Contracts, states and specification:** `docs/contracts/CONTRACTS.md`, `docs/dialogue/STATES.md`, `docs/HANDOVER.md`, `docs/VERTICAL_SLICE.md`.
- **External approvals:** `docs/EXTERNAL_DECISIONS.md`, entries EXT-001 to EXT-129, and the contract changes PC-11 and PC-12.
- **Problem statement:** the official SIH26093 record [1]; the same text is quoted in `docs/HANDOVER.md`.
- **Projections:** derived in §9.1 from the published anchors cited in Table 27 and the local measurements in Tables 17, 16 and 18, under the configuration in Table 20.

B Glossary

Term	Meaning
NHAA 14566	National Helpline Against Atrocities, the toll-free helpline this module serves
SVI	Stress Vulnerability Index: 0–100, nine weighted dimensions, evidence-linked
Band	Low, Moderate, High or Critical
D1–D9	The nine assessment dimensions (Table 7)
SX	The crisis interrupt state: fixed script, Critical, human takeover, no resumption
SH	Human handoff state
Abstention	Returning <i>Needs Human Assessment</i> instead of a score
Escalation packet	The decision-ready case bundle handed to an executive
Licensed question	The single question a dialogue state permits
Shadow mode	A model whose output is displayed beside the authoritative pipeline but can change nothing
UAR	Unweighted average recall: the mean of per-class recall, so every class counts equally (chance is 0.20 for five classes)
AUROC	Area under the ROC curve: how well a score ranks positives above negatives (0.5 is chance)
Weak supervision	Training on a borrowed source label, for example a subreddit of origin, that is not a SAHAY judgement, with its caveat recorded
Exposed fixture	A test case whose result was published during development; its later results are regression evidence, not generalisation
Evidence class	The label stating whether a number is synthetic, contaminated regression, or official (none of ours are official)
DPDP	Digital Personal Data Protection Act 2023 and its 2025 Rules
DLT	The distributed-ledger registration required for commercial voice and SMS traffic in India

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